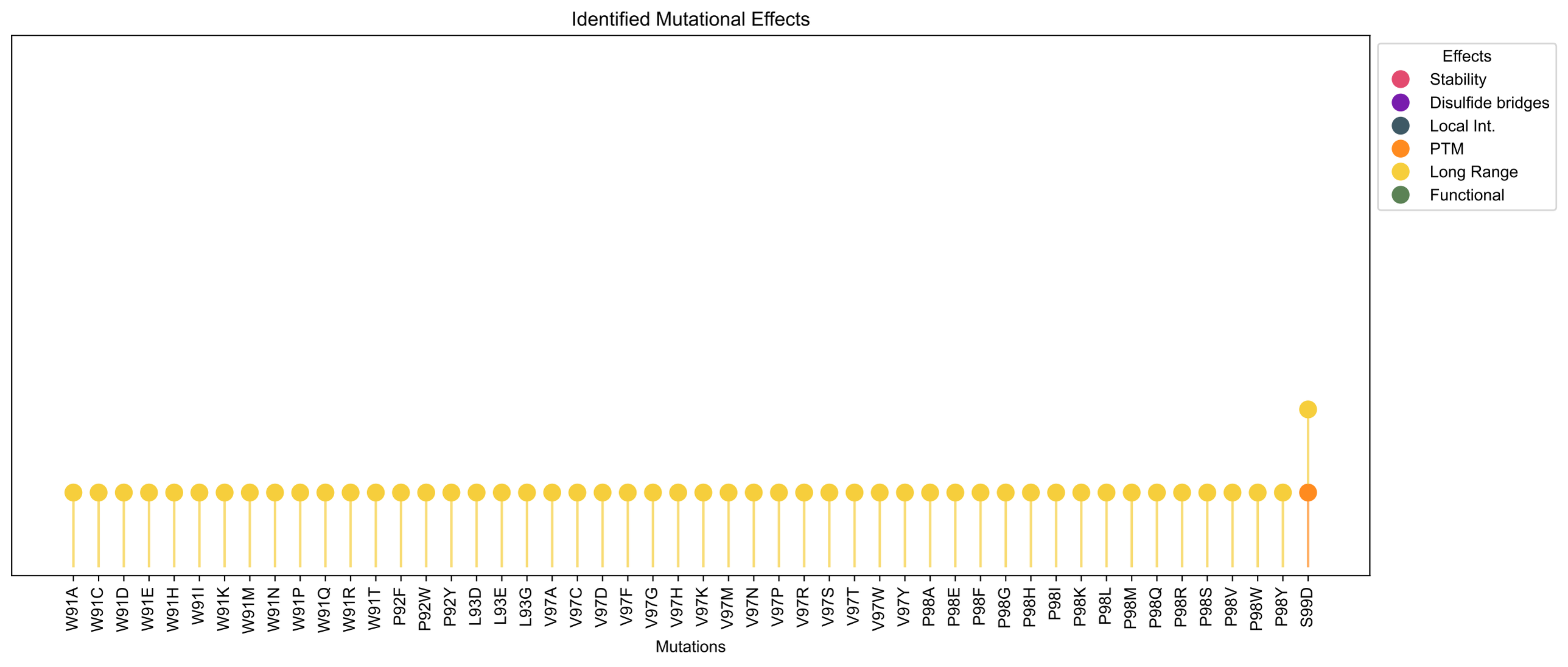
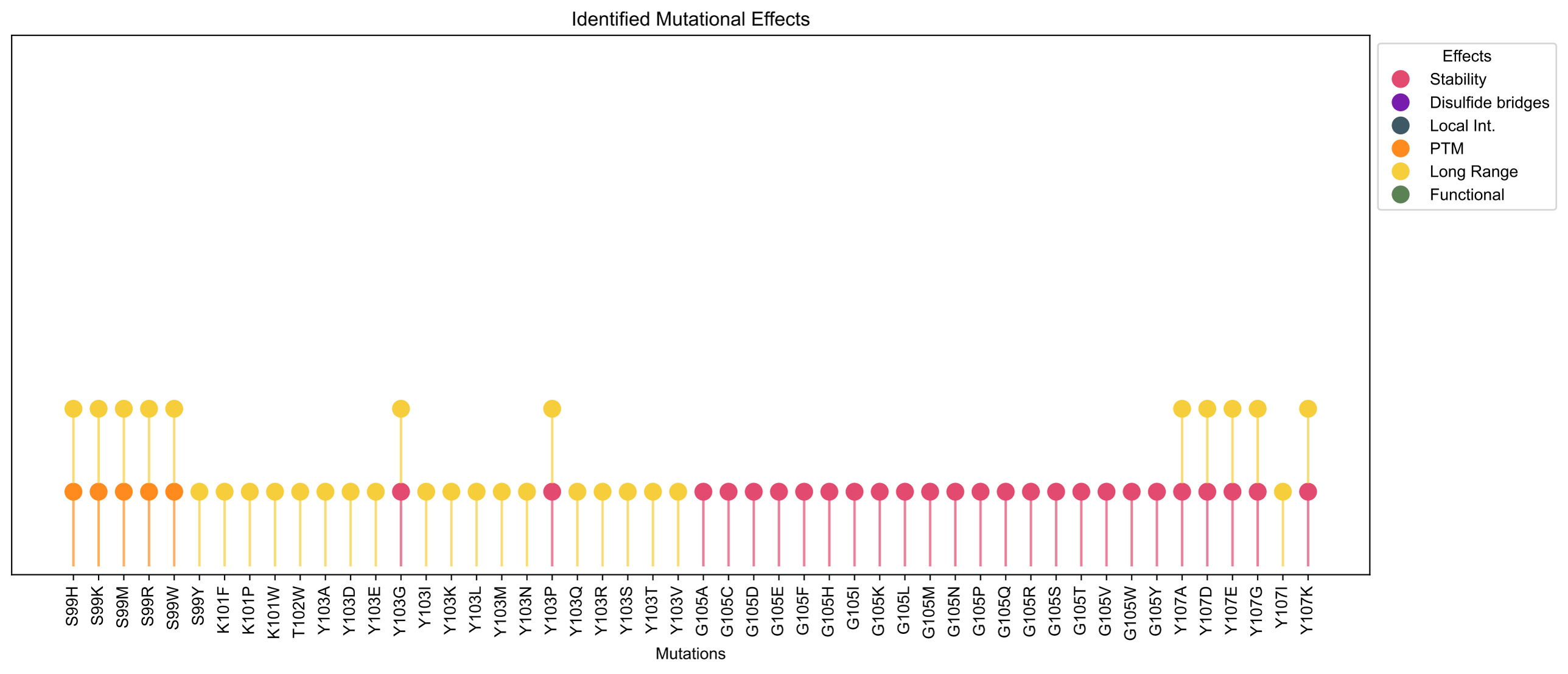
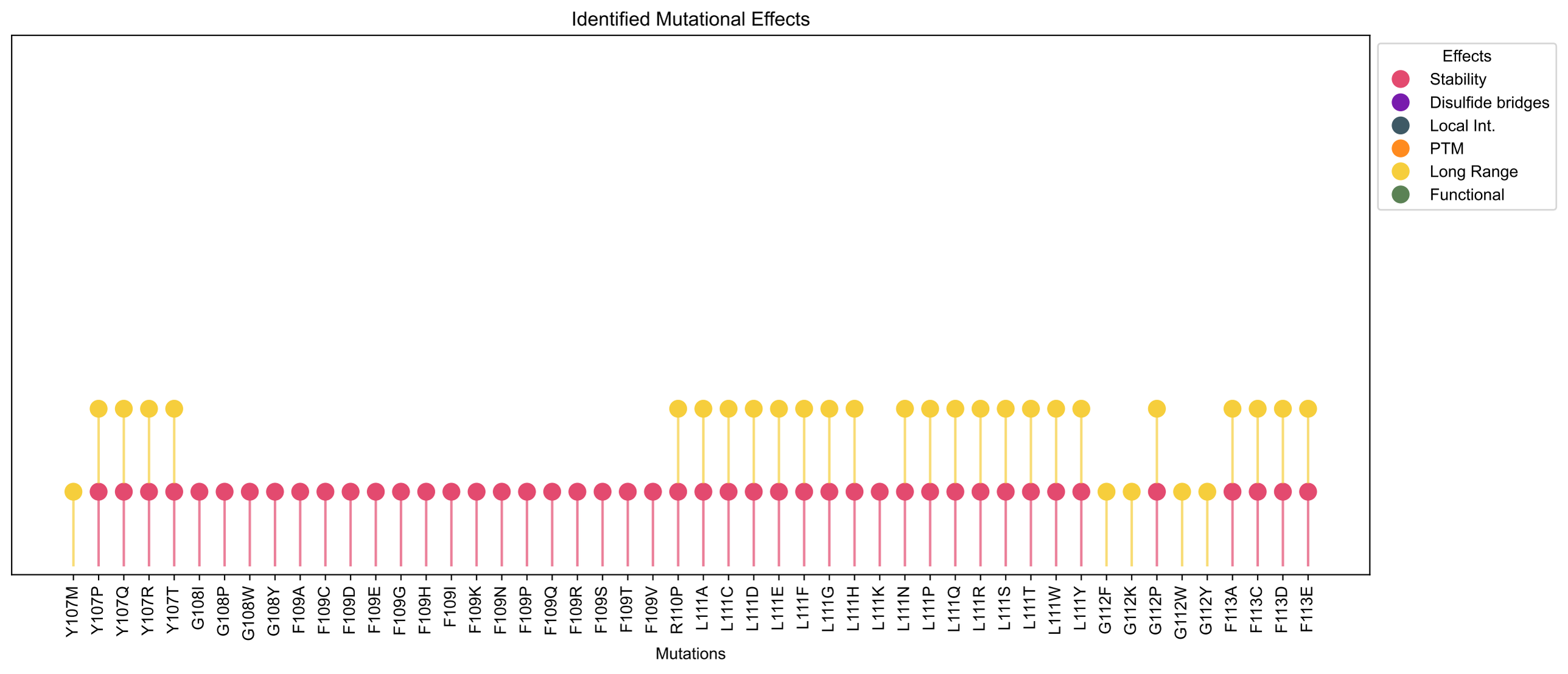
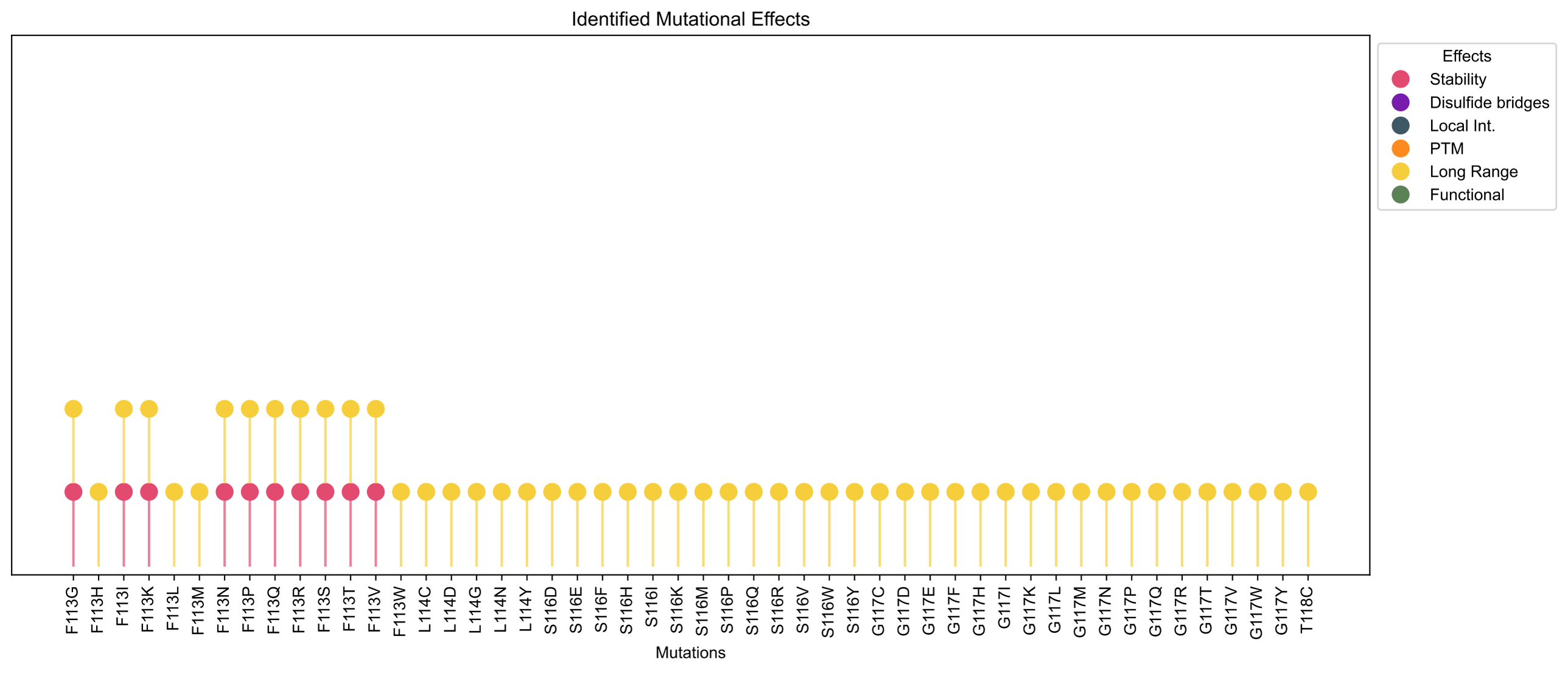
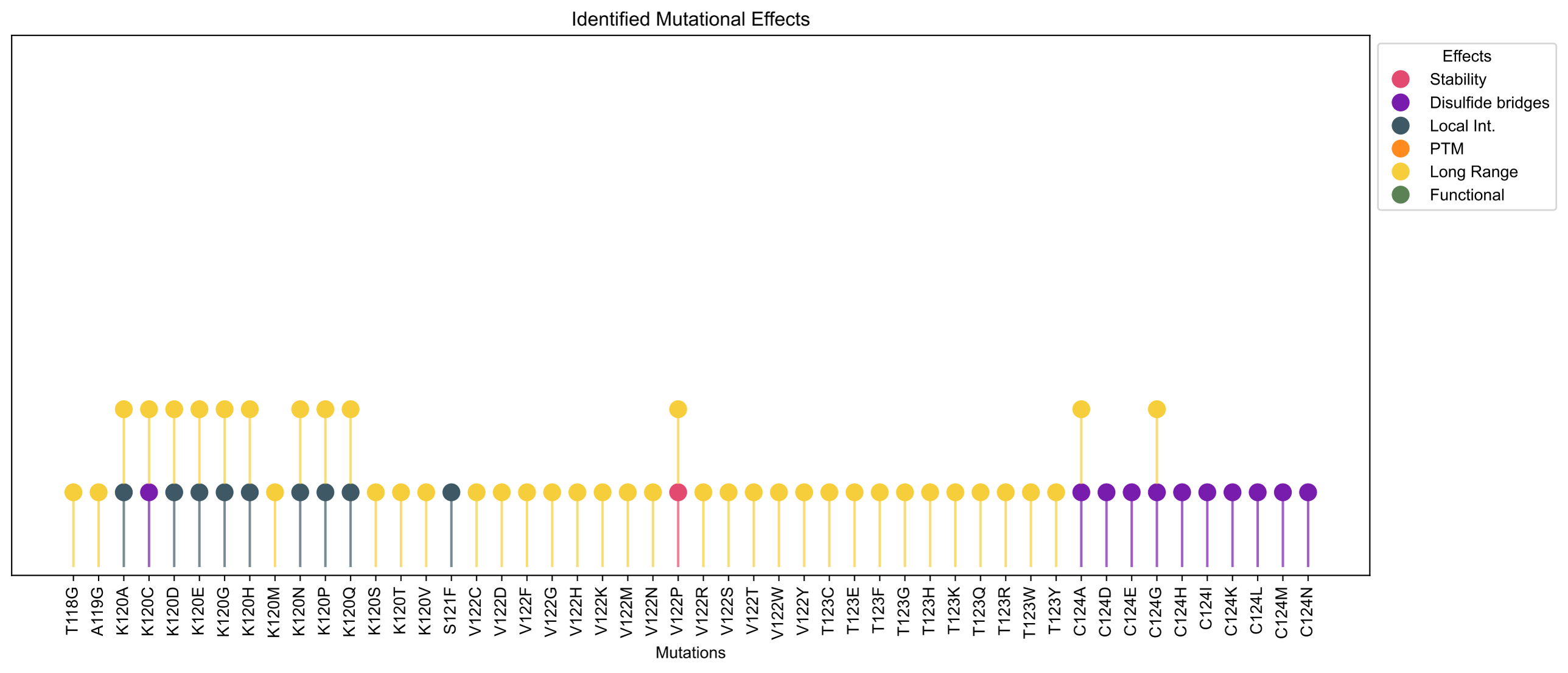
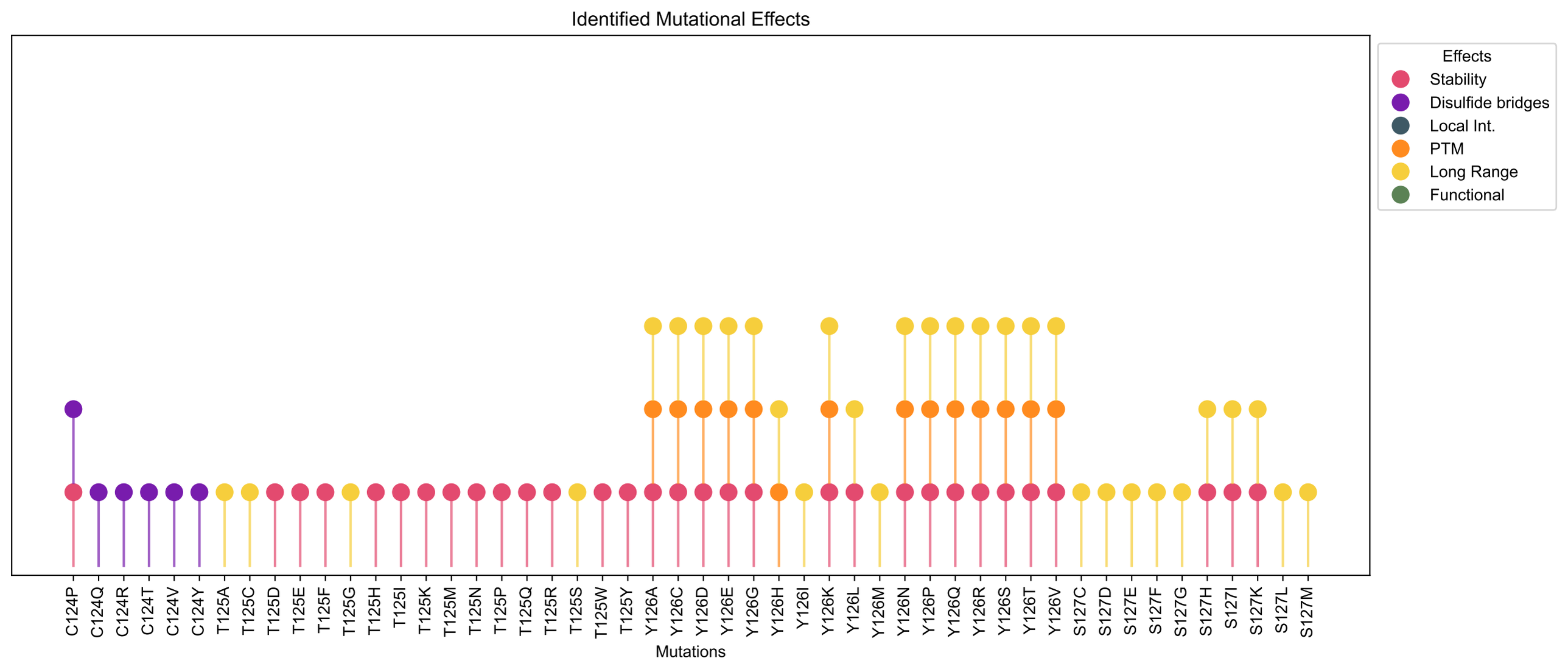

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
W91A
W91C
W91D
W91E
W91H
W91I
W91K
W91M
W91N
W91P
W91Q
W91R
W91T
P92F
P92W
P92Y
L93D
L93E
L93G
V97A
V97C
V97D
V97F
V97G
V97H
V97K
V97M
V97N
V97P
V97R
V97S
V97T
V97W
V97Y
P98A
P98E
P98F
P98G
P98H
P98I
P98K
P98L
P98M
P98Q
P98R
P98S
P98V
P98W
P98Y
S99D
Mutations

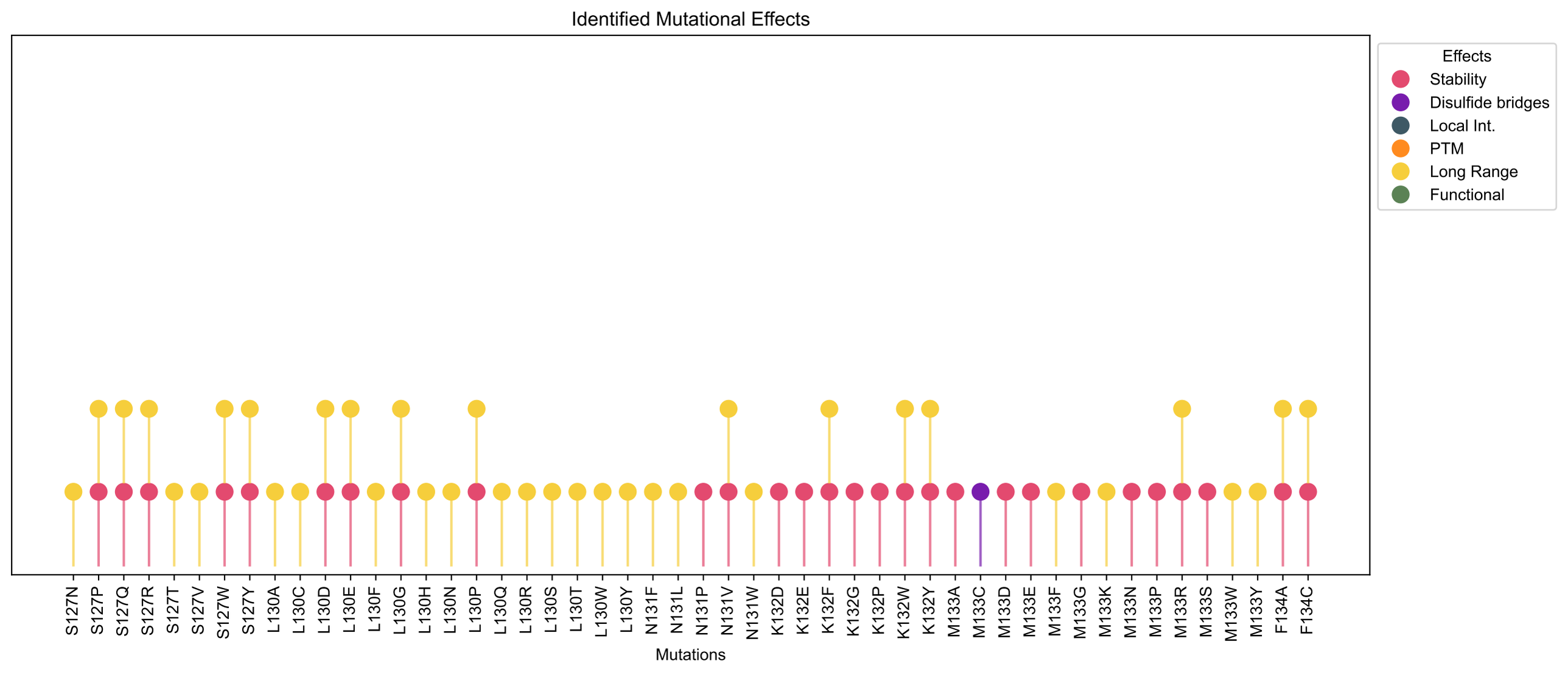

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
S99H
S99K
S99M
S99R
S99W
S99Y
K101F
K101P
K101W
T102W
Y103A
Y103D
Y103E
Y103G
Y103I
Y103K
Y103L
Y103M
Y103N
Y103P
Y103Q
Y103R
Y103S
Y103T
Y103V
G105A
G105C
G105D
G105E
G105F
G105H
G105I
G105K
G105L
G105M
G105N
G105P
G105Q
G105R
G105S
G105T
G105V
G105W
G105Y
Y107A
Y107D
Y107E
Y107G
Y107I
Y107K
Mutations

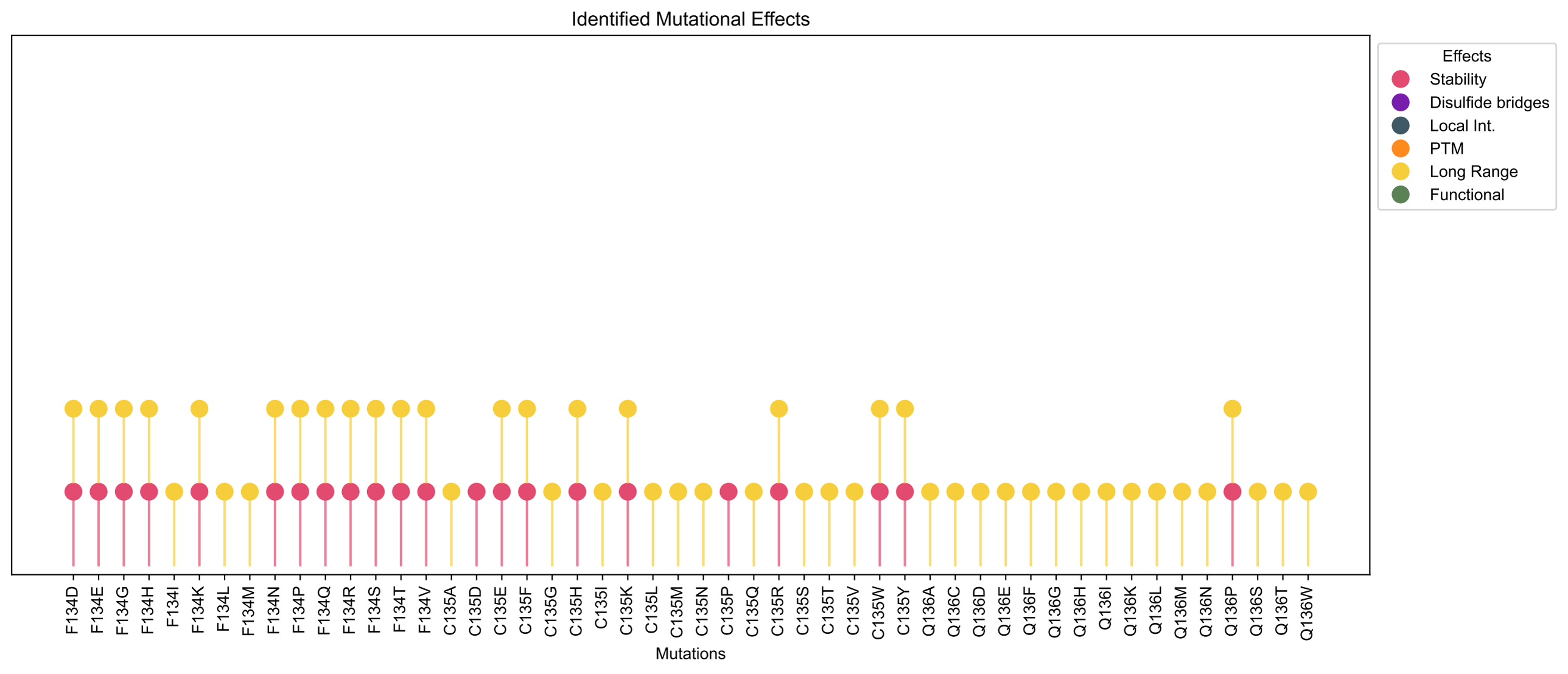

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
Y107M
Y107P
Y107Q
Y107R
Y107T
G108I
G108P
G108W
G108Y
F109A
F109C
F109D
F109E
F109G
F109H
F109I
F109K
F109N
F109P
F109Q
F109R
F109S
F109T
F109V
R110P
L111A
L111C
L111D
L111E
L111F
L111G
L111H
L111K
L111N
L111P
L111Q
L111R
L111S
L111T
L111W
L111Y
G112F
G112K
G112P
G112W
G112Y
F113A
F113C
F113D
F113E
Mutations

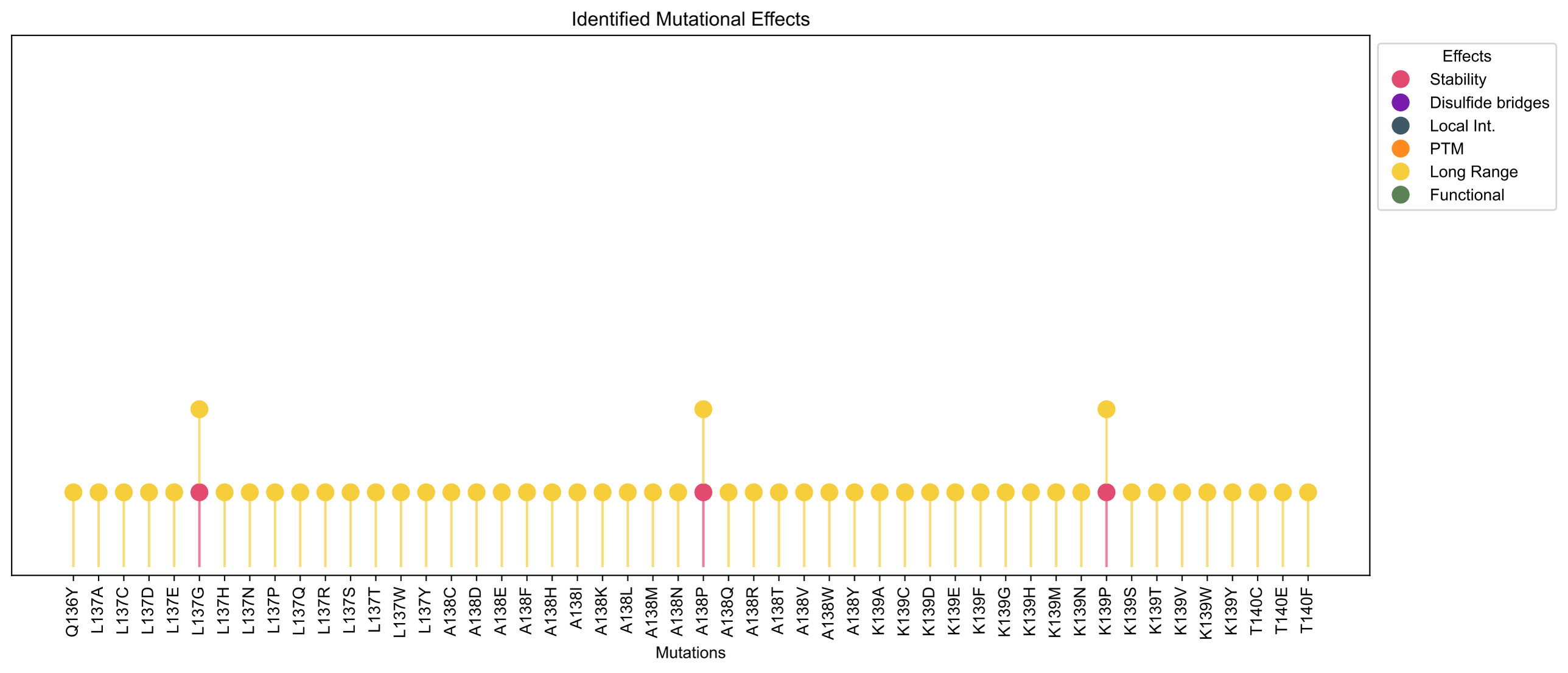

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
F113G
F113H
F113I
F113K
F113L
F113M
F113N
F113P
F113Q
F113R
F113S
F113T
F113V
F113W
L114C
L114D
L114G
L114N
L114Y
S116D
S116E
S116F
S116H
S116I
S116K
S116M
S116P
S116Q
S116R
S116V
S116W
S116Y
G117C
G117D
G117E
G117F
G117H
G117I
G117K
G117L
G117M
G117N
G117P
G117Q
G117R
G117T
G117V
G117W
G117Y
T118C
Mutations

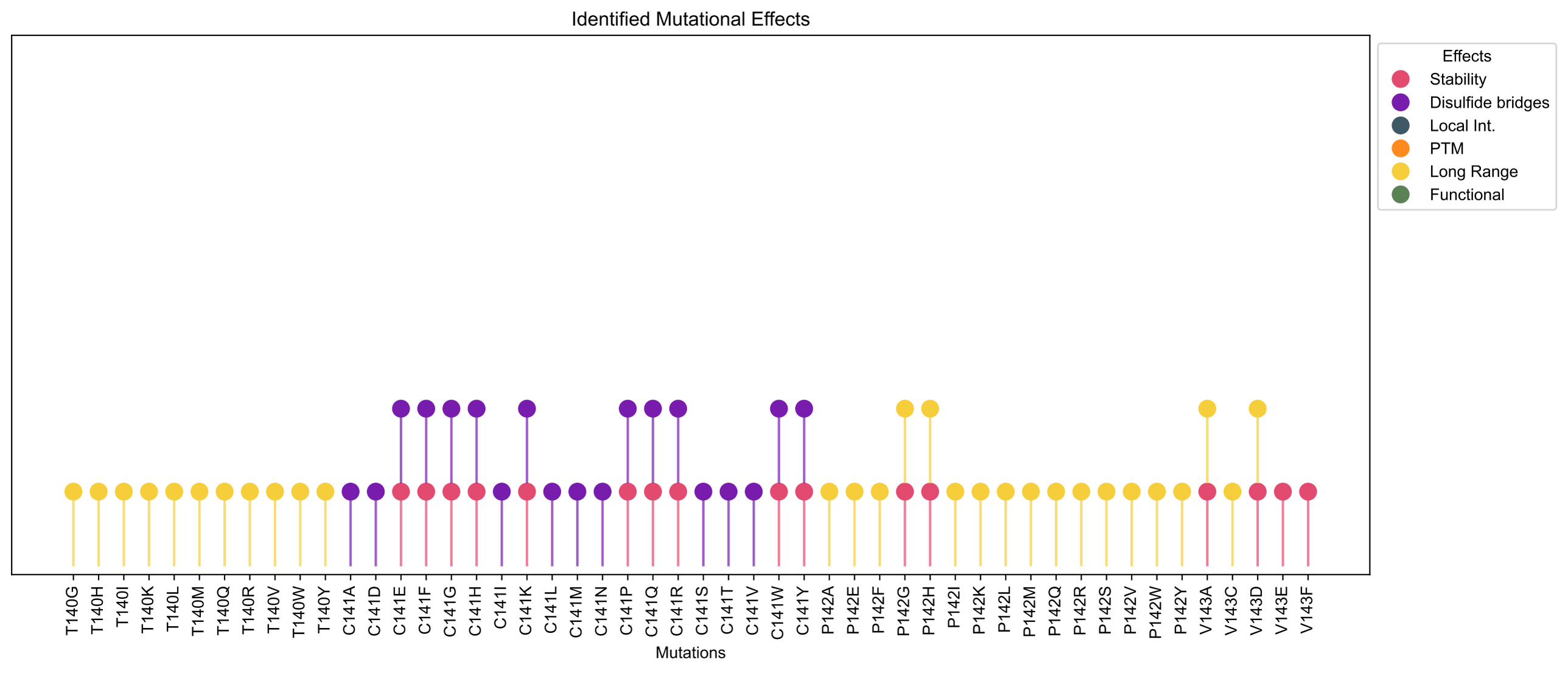

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
T118G
A119G
K120A
K120C
K120D
K120E
K120G
K120H
K120M
K120N
K120P
K120Q
K120S
K120T
K120V
S121F
V122C
V122D
V122F
V122G
V122H
V122K
V122M
V122N
V122P
V122R
V122S
V122T
V122W
V122Y
T123C
T123E
T123F
T123G
T123H
T123K
T123Q
T123R
T123W
T123Y
C124A
C124D
C124E
C124G
C124H
C124I
C124K
C124L
C124M
C124N
Mutations

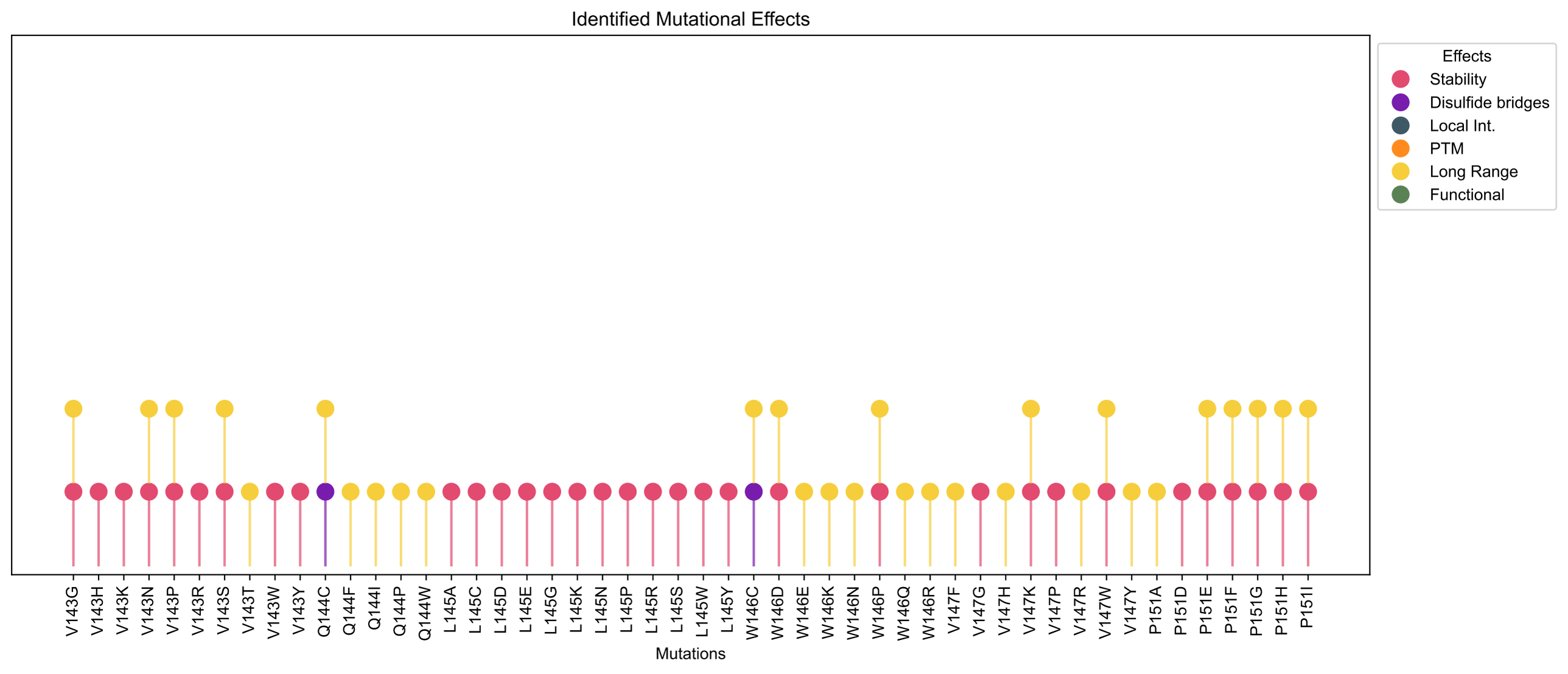

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
C124P
C124Q
C124R
C124T
C124V
C124Y
T125A
T125C
T125D
T125E
T125F
T125G
T125H
T125I
T125K
T125M
T125N
T125P
T125Q
T125R
T125S
T125W
T125Y
Y126A
Y126C
Y126D
Y126E
Y126G
Y126H
Y126I
Y126K
Y126L
Y126M
Y126N
Y126P
Y126Q
Y126R
Y126S
Y126T
Y126V
S127C
S127D
S127E
S127F
S127G
S127H
S127I
S127K
S127L
S127M
Mutations

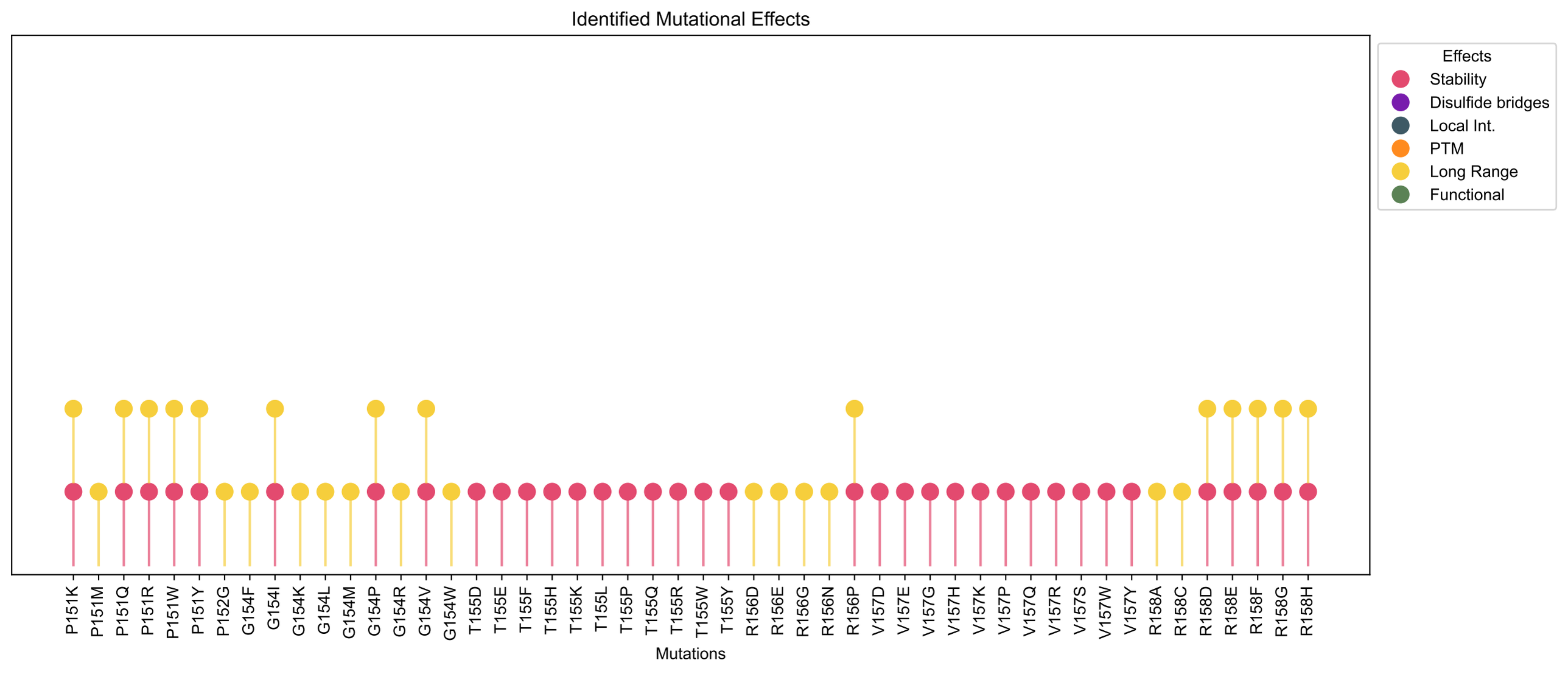

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
S127N
S127P
S127Q
S127R
S127T
S127V
S127W
S127Y
L130A
L130C
L130D
L130E
L130F
L130G
L130H
L130N
L130P
L130Q
L130R
L130S
L130T
L130W
L130Y
N131F
N131L
N131P
N131V
N131W
K132D
K132E
K132F
K132G
K132P
K132W
K132Y
M133A
M133C
M133D
M133E
M133F
M133G
M133K
M133N
M133P
M133R
M133S
M133W
M133Y
F134A
F134C
Mutations

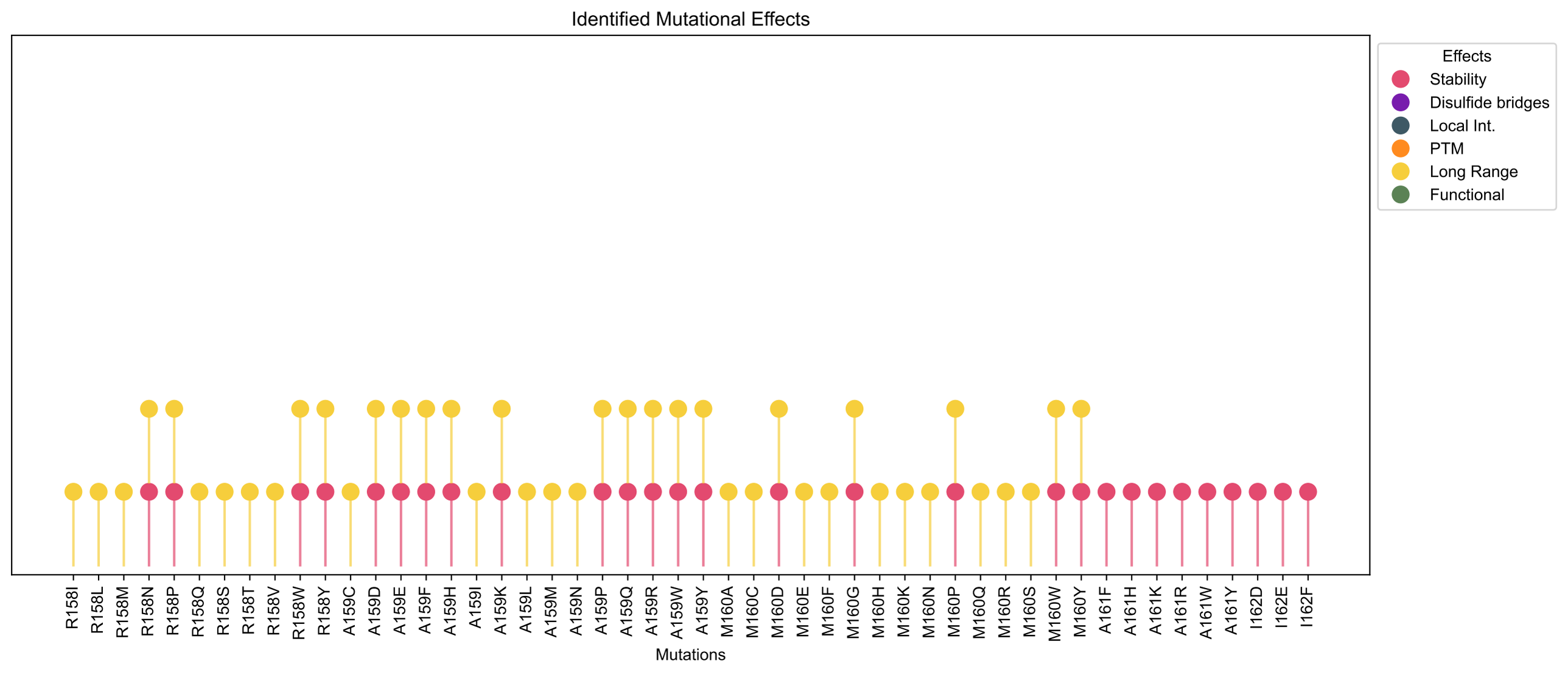

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
F134D
F134E
F134G
F134H
F134I
F134K
F134L
F134M
F134N
F134P
F134Q
F134R
F134S
F134T
F134V
C135A
C135D
C135E
C135F
C135G
C135H
C135I
C135K
C135L
C135M
C135N
C135P
C135Q
C135R
C135S
C135T
C135V
C135W
C135Y
Q136A
Q136C
Q136D
Q136E
Q136F
Q136G
Q136H
Q136I
Q136K
Q136L
Q136M
Q136N
Q136P
Q136S
Q136T
Q136W
Mutations

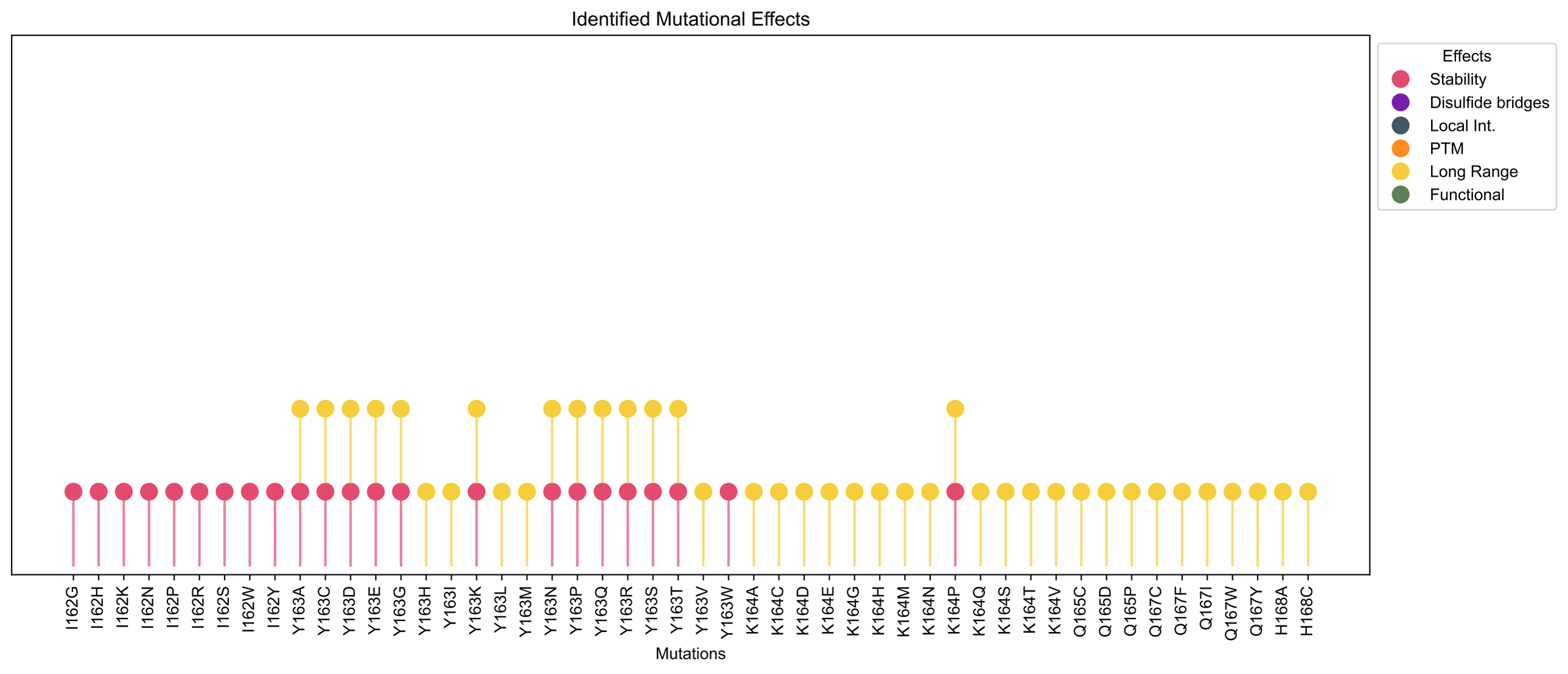

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
Q136Y
L137A
L137C
L137D
L137E
L137G
L137H
L137N
L137P
L137Q
L137R
L137S
L137T
L137W
L137Y
A138C
A138D
A138E
A138F
A138H
A138I
A138K
A138L
A138M
A138N
A138P
A138Q
A138R
A138T
A138V
A138W
A138Y
K139A
K139C
K139D
K139E
K139F
K139G
K139H
K139M
K139N
K139P
K139S
K139T
K139V
K139W
K139Y
T140C
T140E
T140F
Mutations

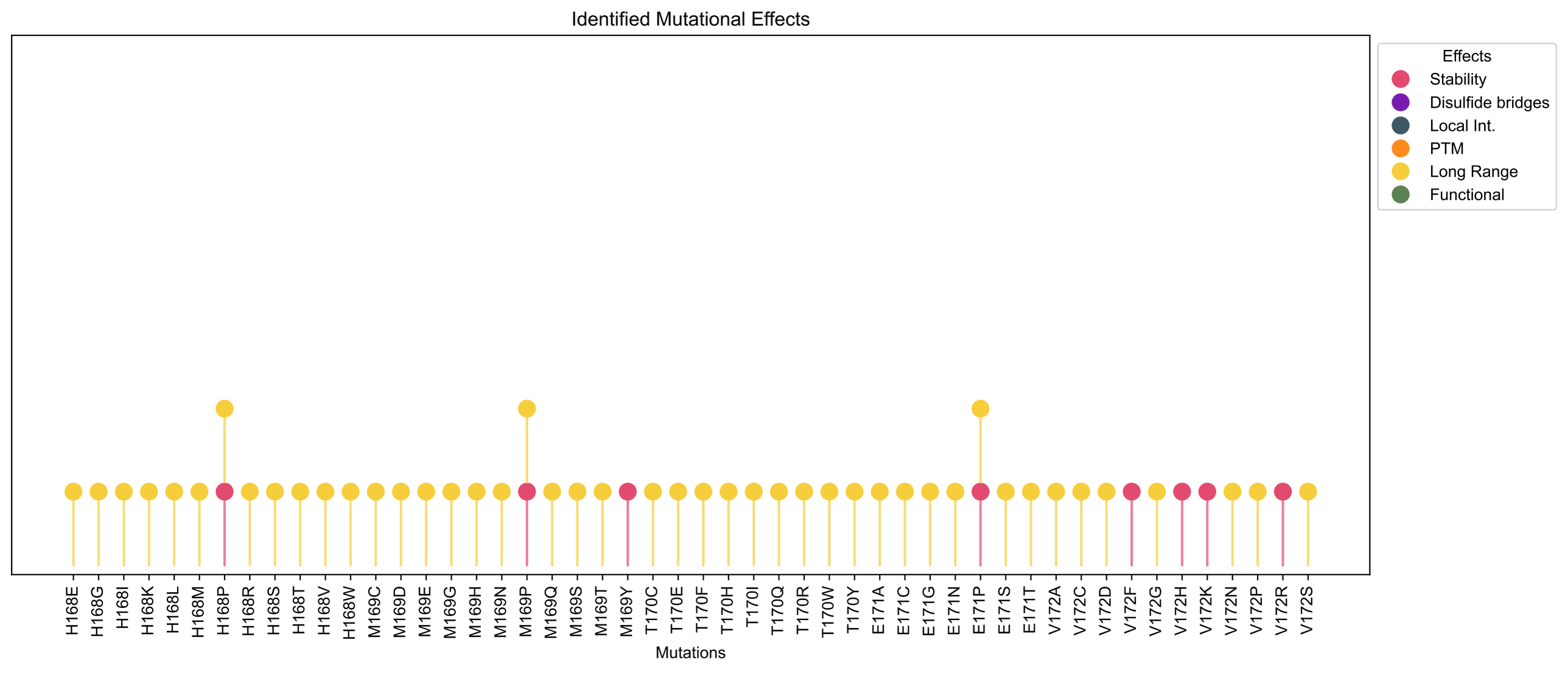

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
T140G
T140H
T140I
T140K
T140L
T140M
T140Q
T140R
T140V
T140W
T140Y
C141A
C141D
C141E
C141F
C141G
C141H
C141I
C141K
C141L
C141M
C141N
C141P
C141Q
C141R
C141S
C141T
C141V
C141W
C141Y
P142A
P142E
P142F
P142G
P142H
P142I
P142K
P142L
P142M
P142Q
P142R
P142S
P142V
P142W
P142Y
V143A
V143C
V143D
V143E
V143F
Mutations

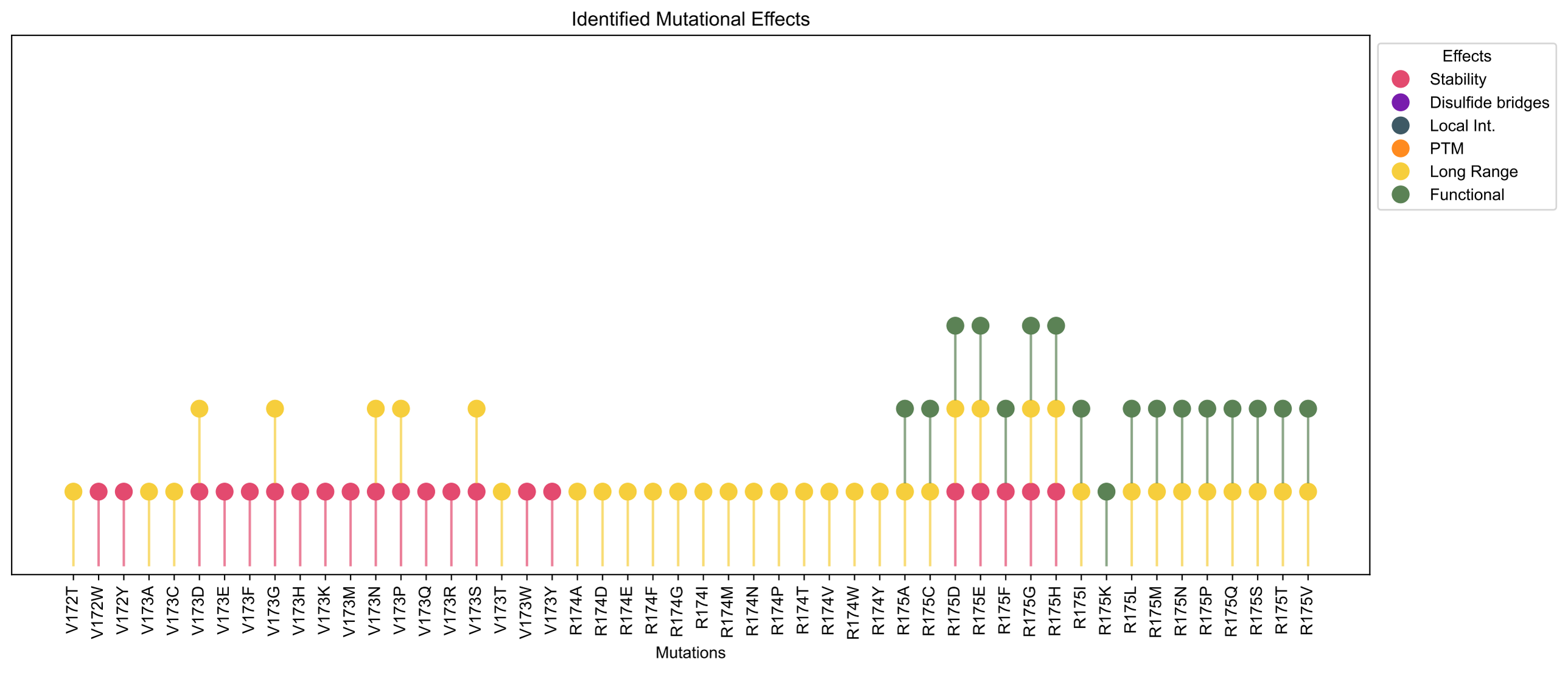

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
V143G
V143H
V143K
V143N
V143P
V143R
V143S
V143T
V143W
V143Y
Q144C
Q144F
Q144I
Q144P
Q144W
L145A
L145C
L145D
L145E
L145G
L145K
L145N
L145P
L145R
L145S
L145W
L145Y
W146C
W146D
W146E
W146K
W146N
W146P
W146Q
W146R
V147F
V147G
V147H
V147K
V147P
V147R
V147W
V147Y
P151A
P151D
P151E
P151F
P151G
P151H
P151I
Mutations

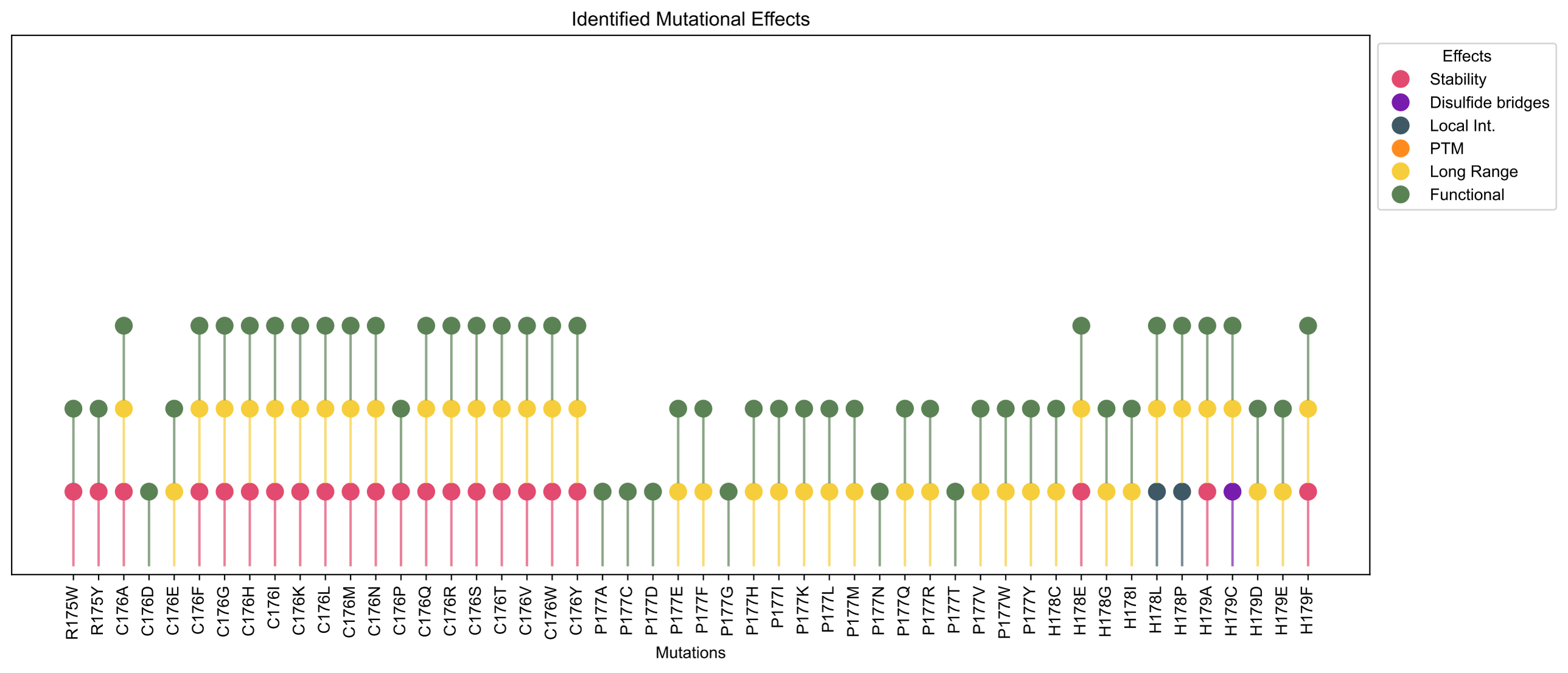

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
P151K
P151M
P151Q
P151R
P151W
P151Y
P152G
G154F
G154I
G154K
G154L
G154M
G154P
G154R
G154V
G154W
T155D
T155E
T155F
T155H
T155K
T155L
T155P
T155Q
T155R
T155W
T155Y
R156D
R156E
R156G
R156N
R156P
V157D
V157E
V157G
V157H
V157K
V157P
V157Q
V157R
V157S
V157W
V157Y
R158A
R158C
R158D
R158E
R158F
R158G
R158H
Mutations

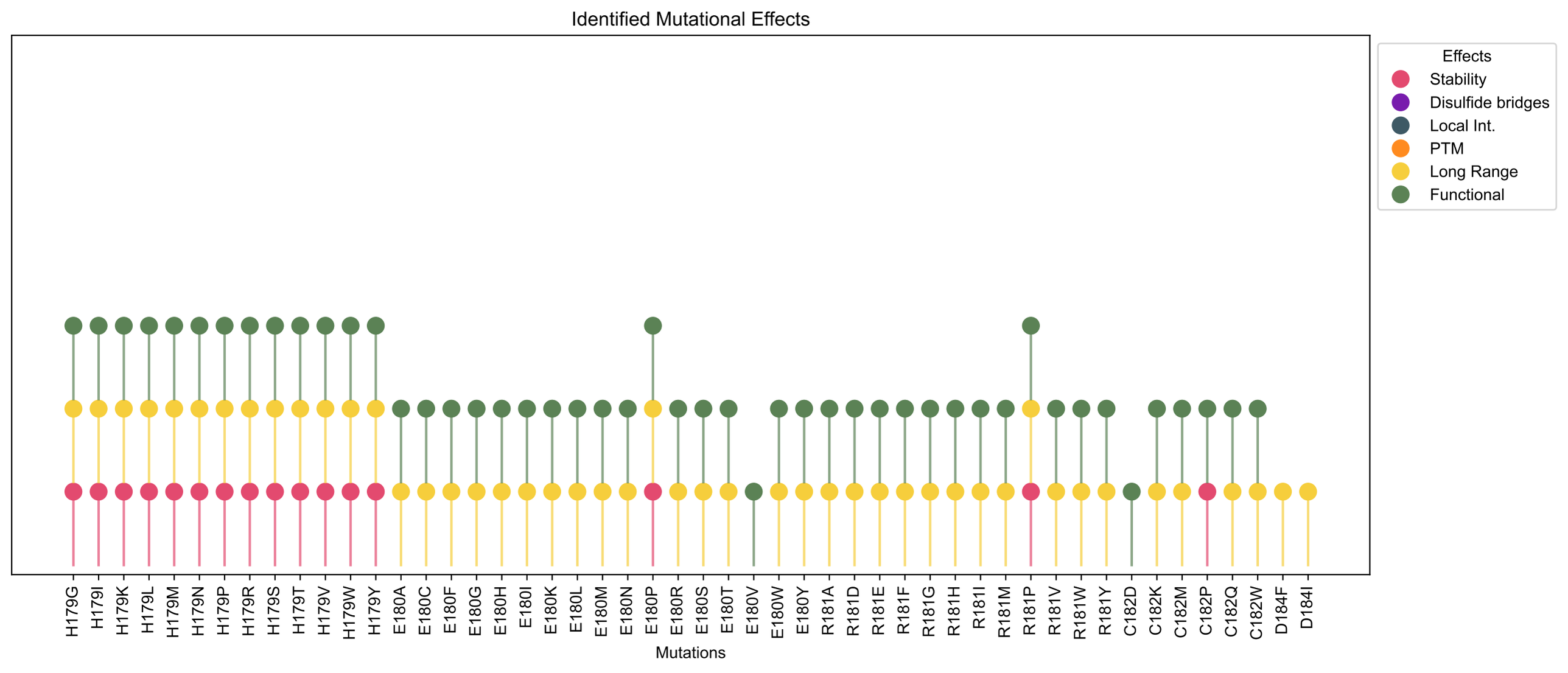

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
R158I
R158L
R158M
R158N
R158P
R158Q
R158S
R158T
R158V
R158W
R158Y
A159C
A159D
A159E
A159F
A159H
A159I
A159K
A159L
A159M
A159N
A159P
A159Q
A159R
A159W
A159Y
M160A
M160C
M160D
M160E
M160F
M160G
M160H
M160K
M160N
M160P
M160Q
M160R
M160S
M160W
M160Y
A161F
A161H
A161K
A161R
A161W
A161Y
I162D
I162E
I162F
Mutations

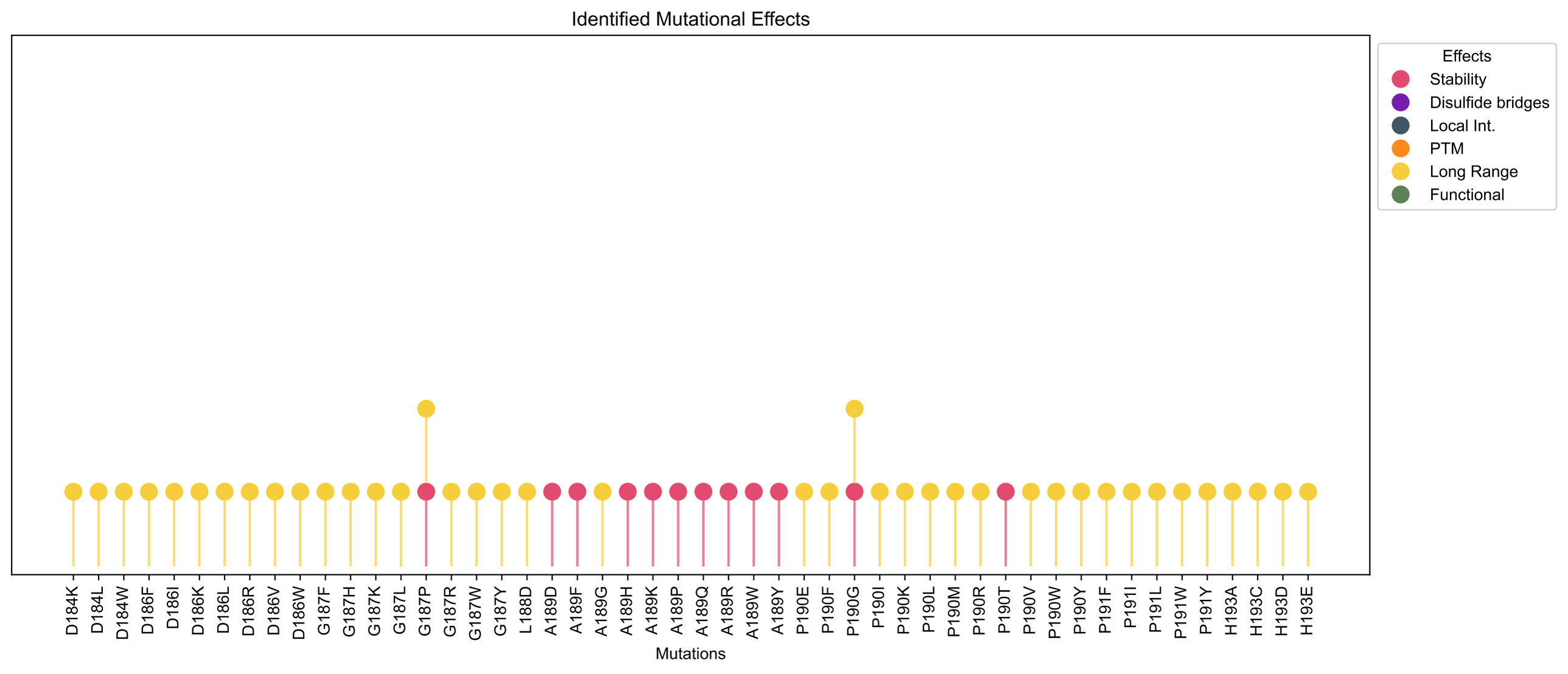

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
I162G
I162H
I162K
I162N
I162P
I162R
I162S
I162W
I162Y
Y163A
Y163C
Y163D
Y163E
Y163G
Y163H
Y163I
Y163K
Y163L
Y163M
Y163N
Y163P
Y163Q
Y163R
Y163S
Y163T
Y163V
Y163W
K164A
K164C
K164D
K164E
K164G
K164H
K164M
K164N
K164P
K164Q
K164S
K164T
K164V
Q165C
Q165D
Q165P
Q167C
Q167F
Q167I
Q167W
Q167Y
H168A
H168C
Mutations

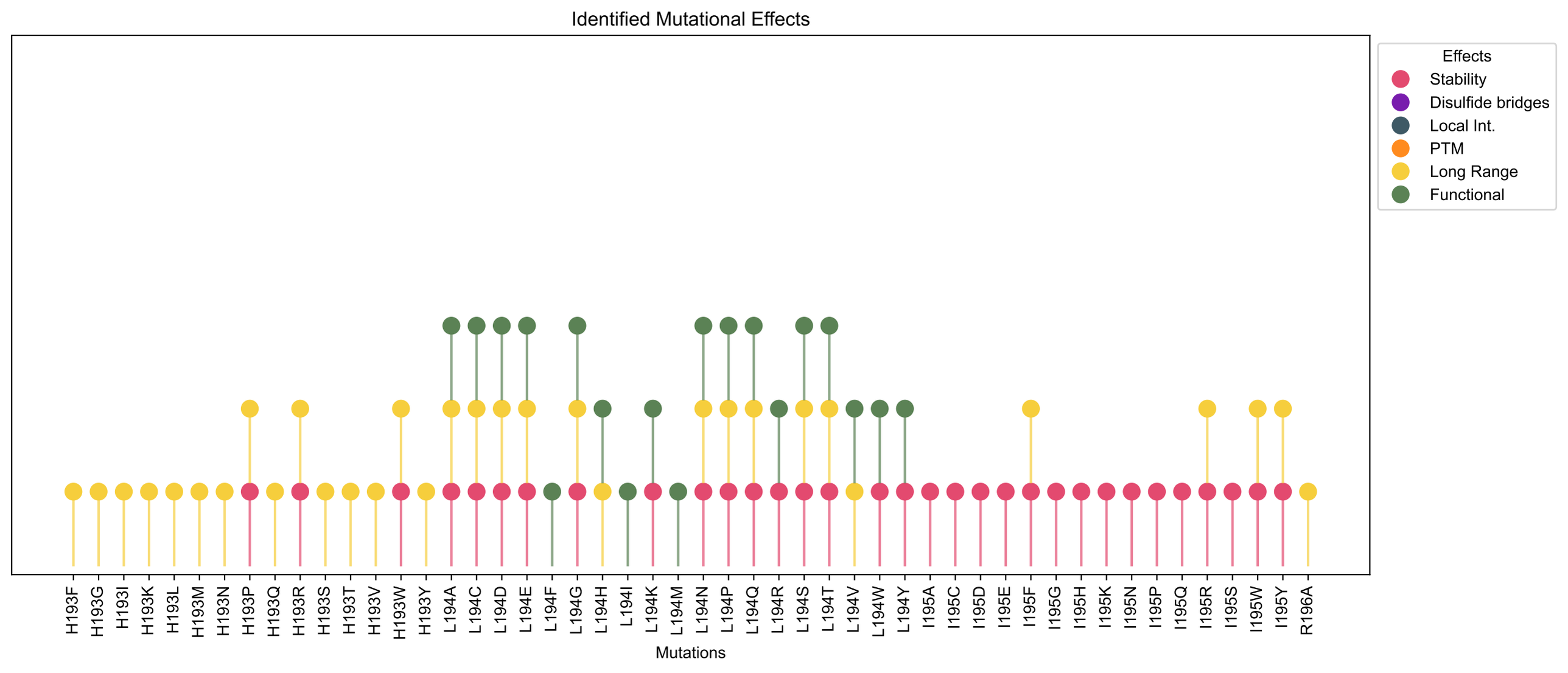

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
H168E
H168G
H168I
H168K
H168L
H168M
H168P
H168R
H168S
H168T
H168V
H168W
M169C
M169D
M169E
M169G
M169H
M169N
M169P
M169Q
M169S
M169T
M169Y
T170C
T170E
T170F
T170H
T170I
T170Q
T170R
T170W
T170Y
E171A
E171C
E171G
E171N
E171P
E171S
E171T
V172A
V172C
V172D
V172F
V172G
V172H
V172K
V172N
V172P
V172R
V172S
Mutations

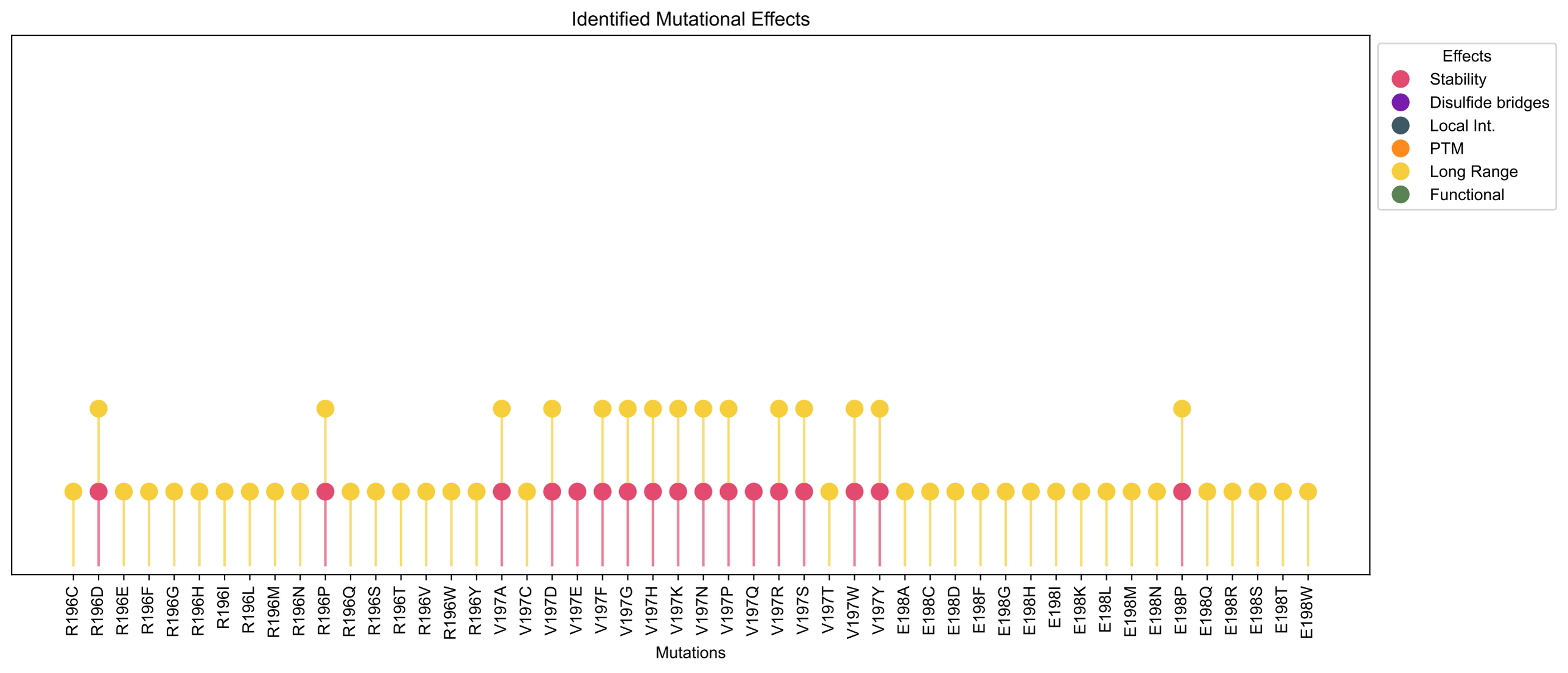

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
V172T
V172W
V172Y
V173A
V173C
V173D
V173E
V173F
V173G
V173H
V173K
V173M
V173N
V173P
V173Q
V173R
V173S
V173T
V173W
V173Y
R174A
R174D
R174E
R174F
R174G
R174I
R174M
R174N
R174P
R174T
R174V
R174W
R174Y
R175A
R175C
R175D
R175E
R175F
R175G
R175H
R175I
R175K
R175L
R175M
R175N
R175P
R175Q
R175S
R175T
R175V
Mutations

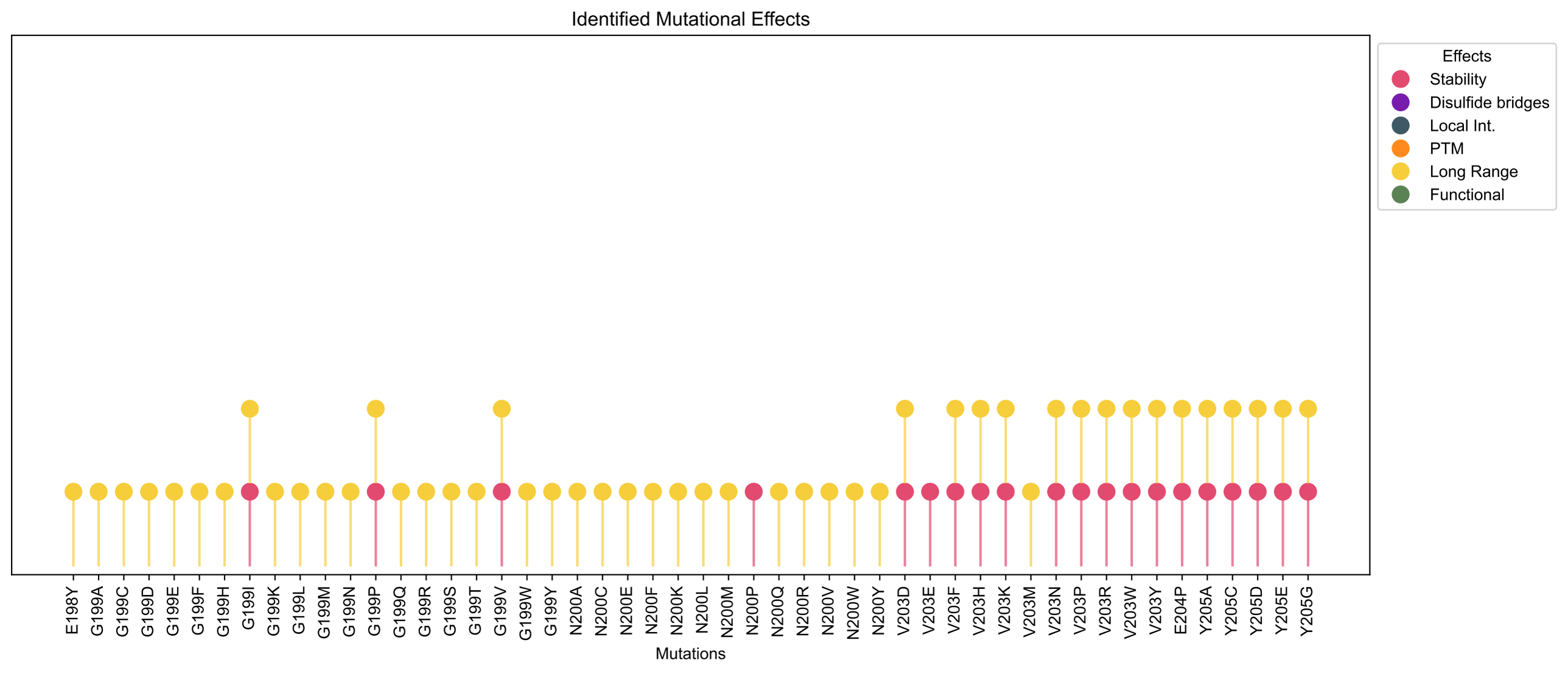

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
R175W
R175Y
C176A
C176D
C176E
C176F
C176G
C176H
C176I
C176K
C176L
C176M
C176N
C176P
C176Q
C176R
C176S
C176T
C176V
C176W
C176Y
P177A
P177C
P177D
P177E
P177F
P177G
P177H
P177I
P177K
P177L
P177M
P177N
P177Q
P177R
P177T
P177V
P177W
P177Y
H178C
H178E
H178G
H178I
H178L
H178P
H179A
H179C
H179D
H179E
H179F
Mutations

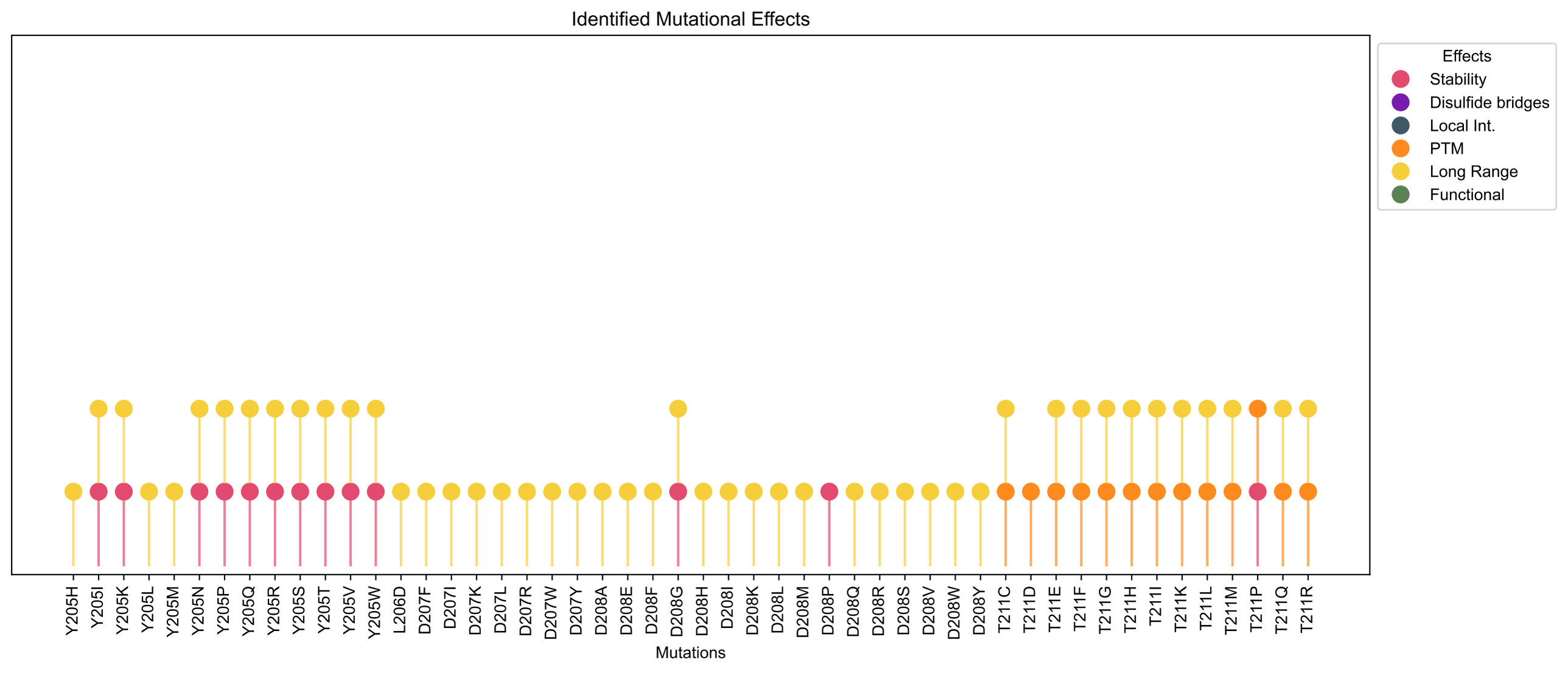

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
H179G
H179I
H179K
H179L
H179M
H179N
H179P
H179R
H179S
H179T
H179V
H179W
H179Y
E180A
E180C
E180F
E180G
E180H
E180I
E180K
E180L
E180M
E180N
E180P
E180R
E180S
E180T
E180V
E180W
E180Y
R181A
R181D
R181E
R181F
R181G
R181H
R181I
R181M
R181P
R181V
R181W
R181Y
C182D
C182K
C182M
C182P
C182Q
C182W
D184F
D184I
Mutations

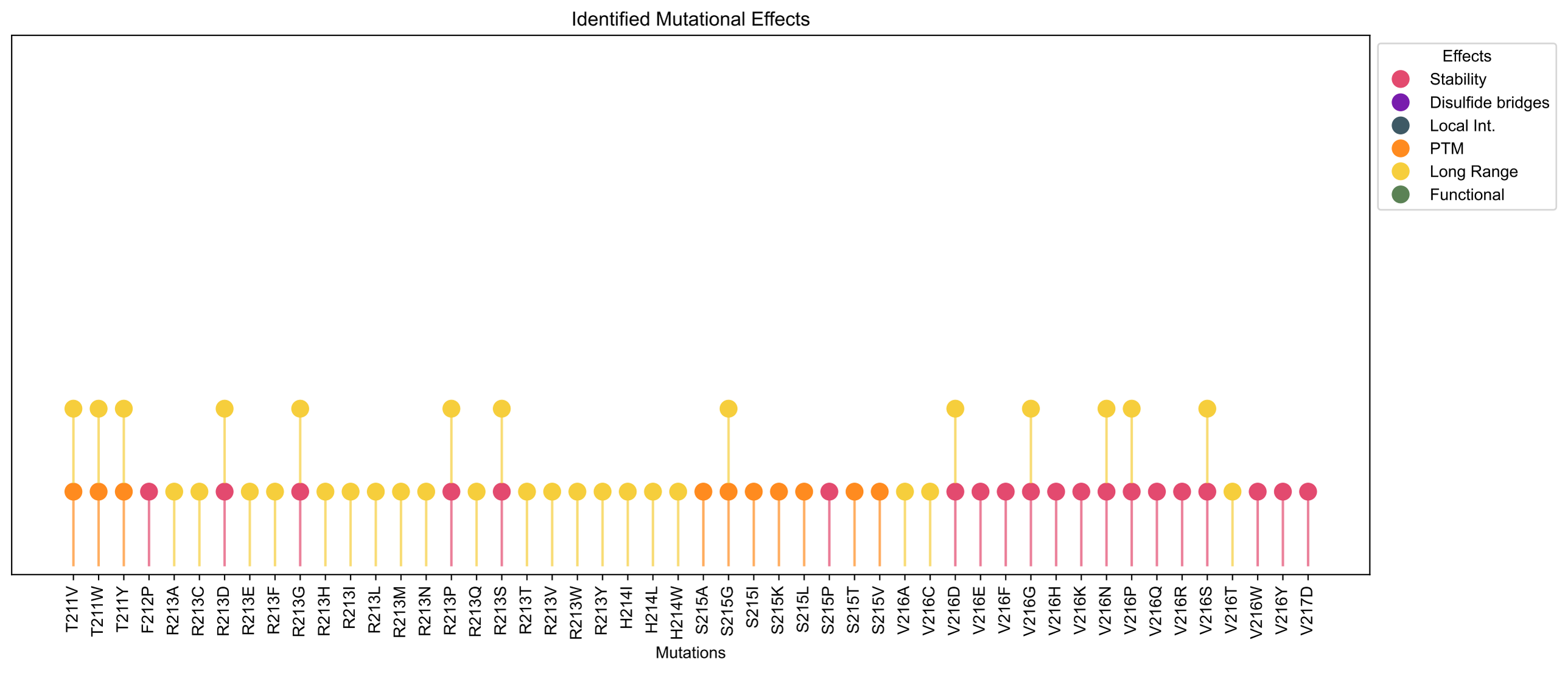

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
D184K
D184L
D184W
D186F
D186I
D186K
D186L
D186R
D186V
D186W
G187F
G187H
G187K
G187L
G187P
G187R
G187W
G187Y
L188D
A189D
A189F
A189G
A189H
A189K
A189P
A189Q
A189R
A189W
A189Y
P190E
P190F
P190G
P190I
P190K
P190L
P190M
P190R
P190T
P190V
P190W
P190Y
P191F
P191I
P191L
P191W
P191Y
H193A
H193C
H193D
H193E
Mutations

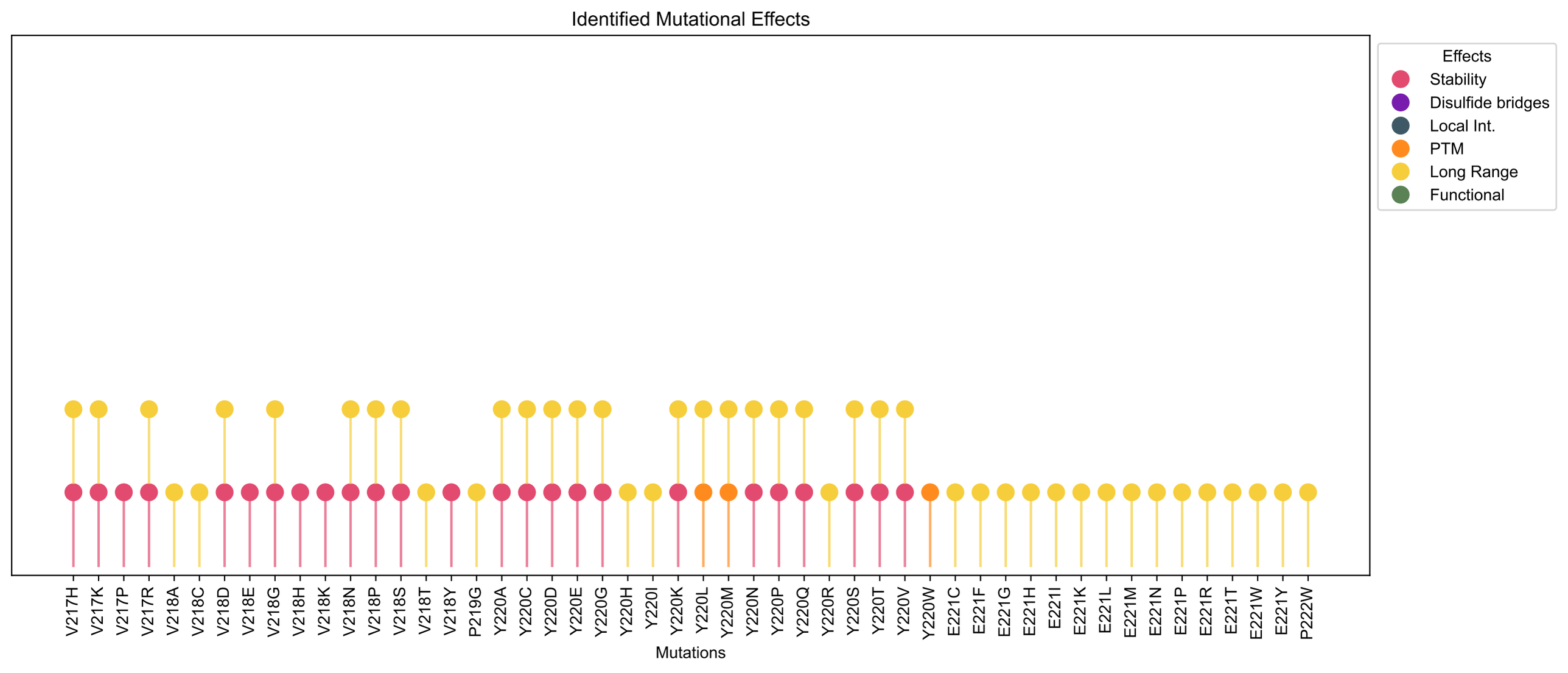

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
H193F
H193G
H193I
H193K
H193L
H193M
H193N
H193P
H193Q
H193R
H193S
H193T
H193V
H193W
H193Y
L194A
L194C
L194D
L194E
L194F
L194G
L194H
L194I
L194K
L194M
L194N
L194P
L194Q
L194R
L194S
L194T
L194V
L194W
L194Y
I195A
I195C
I195D
I195E
I195F
I195G
I195H
I195K
I195N
I195P
I195Q
I195R
I195S
I195W
I195Y
R196A
Mutations

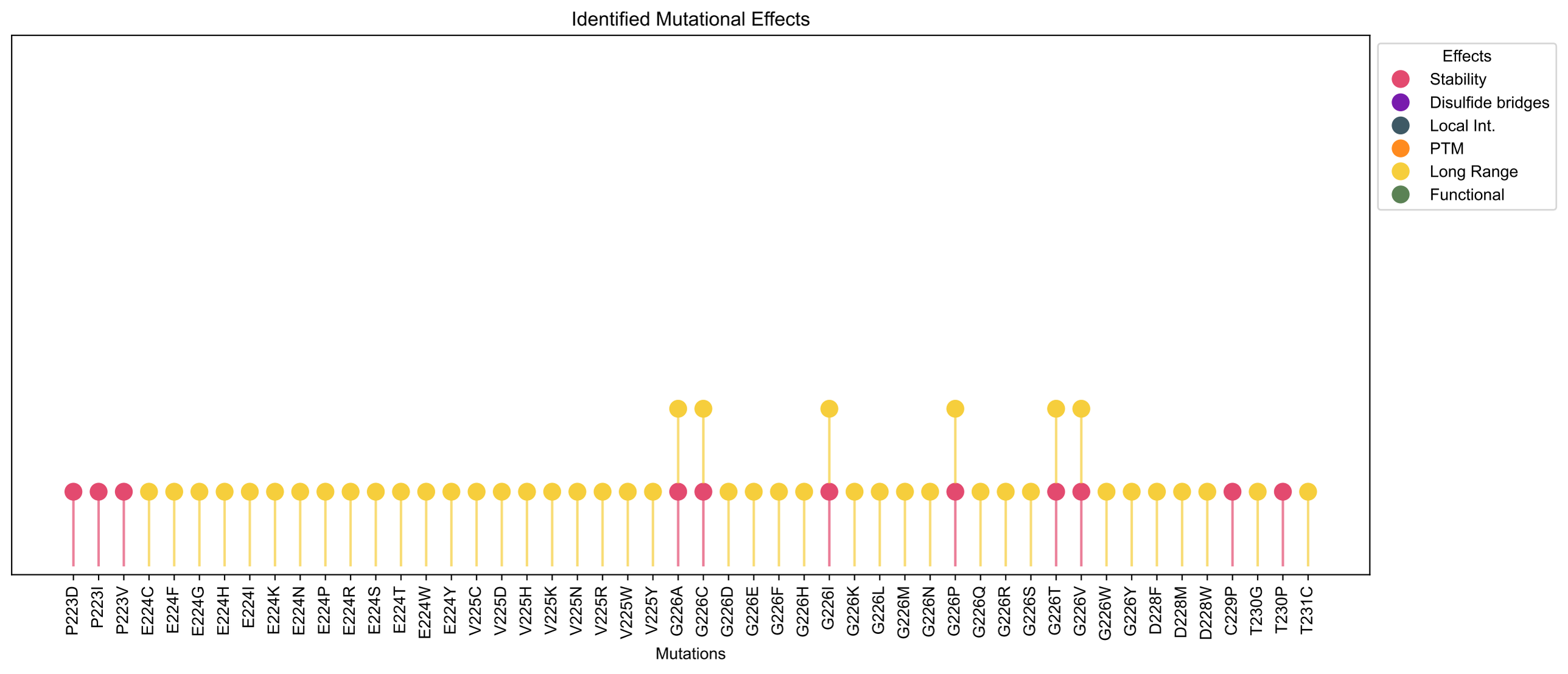

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
R196C
R196D
R196E
R196F
R196G
R196H
R196I
R196L
R196M
R196N
R196P
R196Q
R196S
R196T
R196V
R196W
R196Y
V197A
V197C
V197D
V197E
V197F
V197G
V197H
V197K
V197N
V197P
V197Q
V197R
V197S
V197T
V197W
V197Y
E198A
E198C
E198D
E198F
E198G
E198H
E198I
E198K
E198L
E198M
E198N
E198P
E198Q
E198R
E198S
E198T
E198W
Mutations

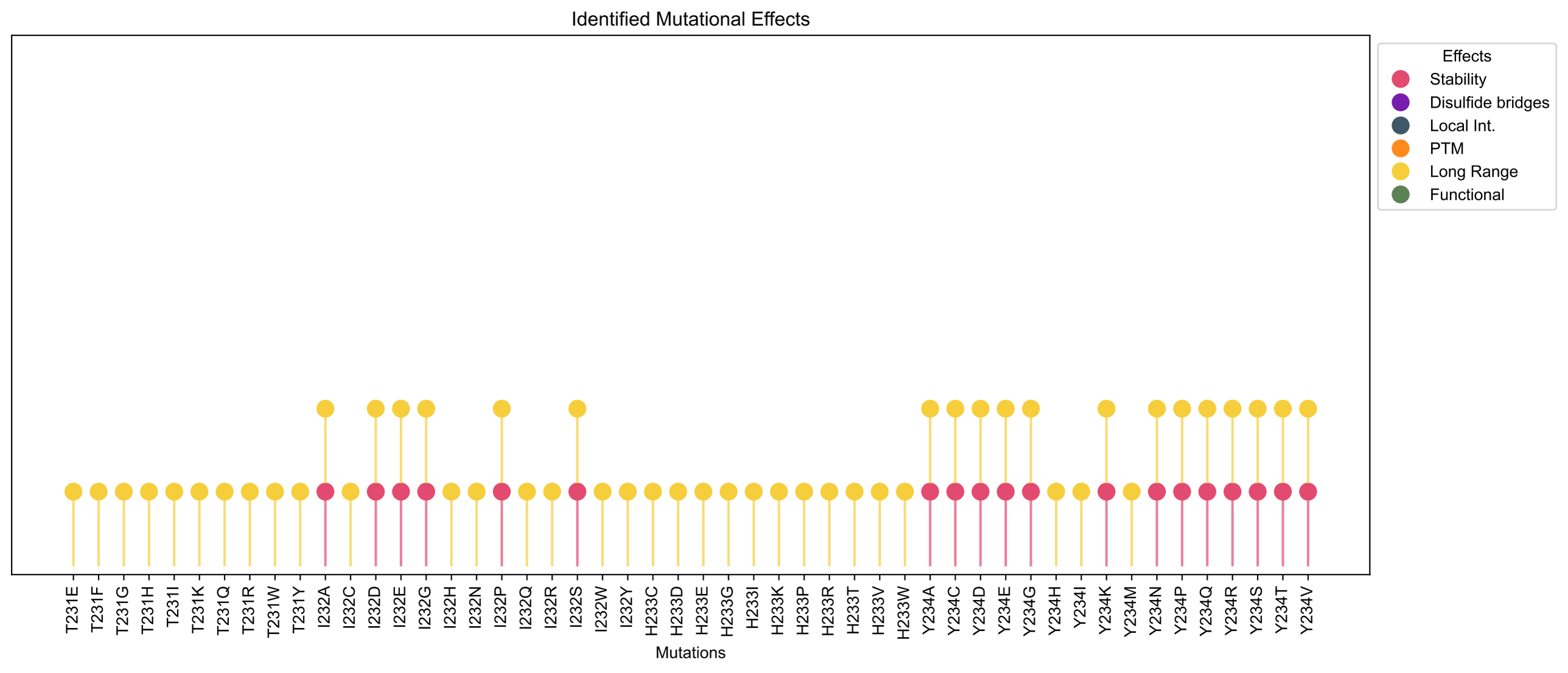

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
E198Y
G199A
G199C
G199D
G199E
G199F
G199H
G199I
G199K
G199L
G199M
G199N
G199P
G199Q
G199R
G199S
G199T
G199V
G199W
G199Y
N200A
N200C
N200E
N200F
N200K
N200L
N200M
N200P
N200Q
N200R
N200V
N200W
N200Y
V203D
V203E
V203F
V203H
V203K
V203M
V203N
V203P
V203R
V203W
V203Y
E204P
Y205A
Y205C
Y205D
Y205E
Y205G
Mutations

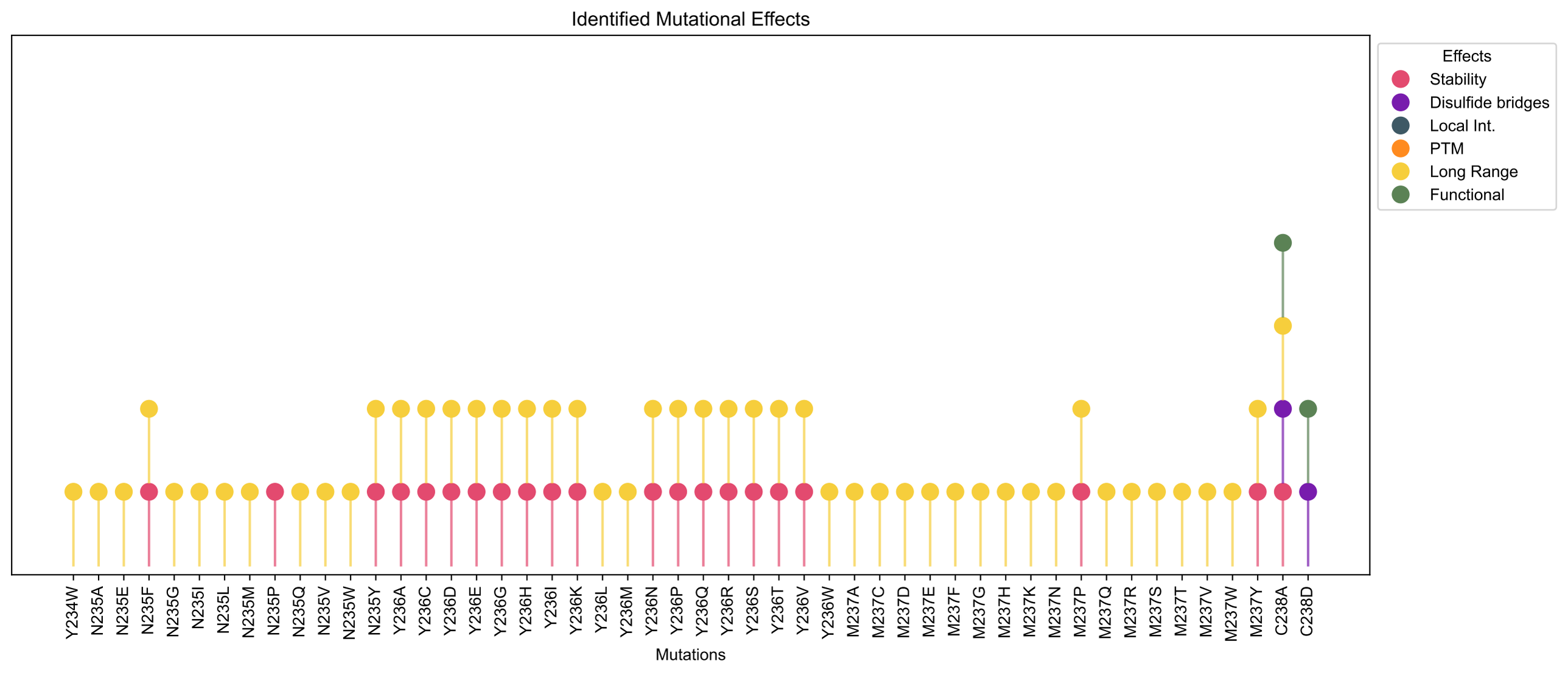

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
Y205H
Y205I
Y205K
Y205L
Y205M
Y205N
Y205P
Y205Q
Y205R
Y205S
Y205T
Y205V
Y205W
L206D
D207F
D207I
D207K
D207L
D207R
D207W
D207Y
D208A
D208E
D208F
D208G
D208H
D208I
D208K
D208L
D208M
D208P
D208Q
D208R
D208S
D208V
D208W
D208Y
T211C
T211D
T211E
T211F
T211G
T211H
T211I
T211K
T211L
T211M
T211P
T211Q
T211R
Mutations

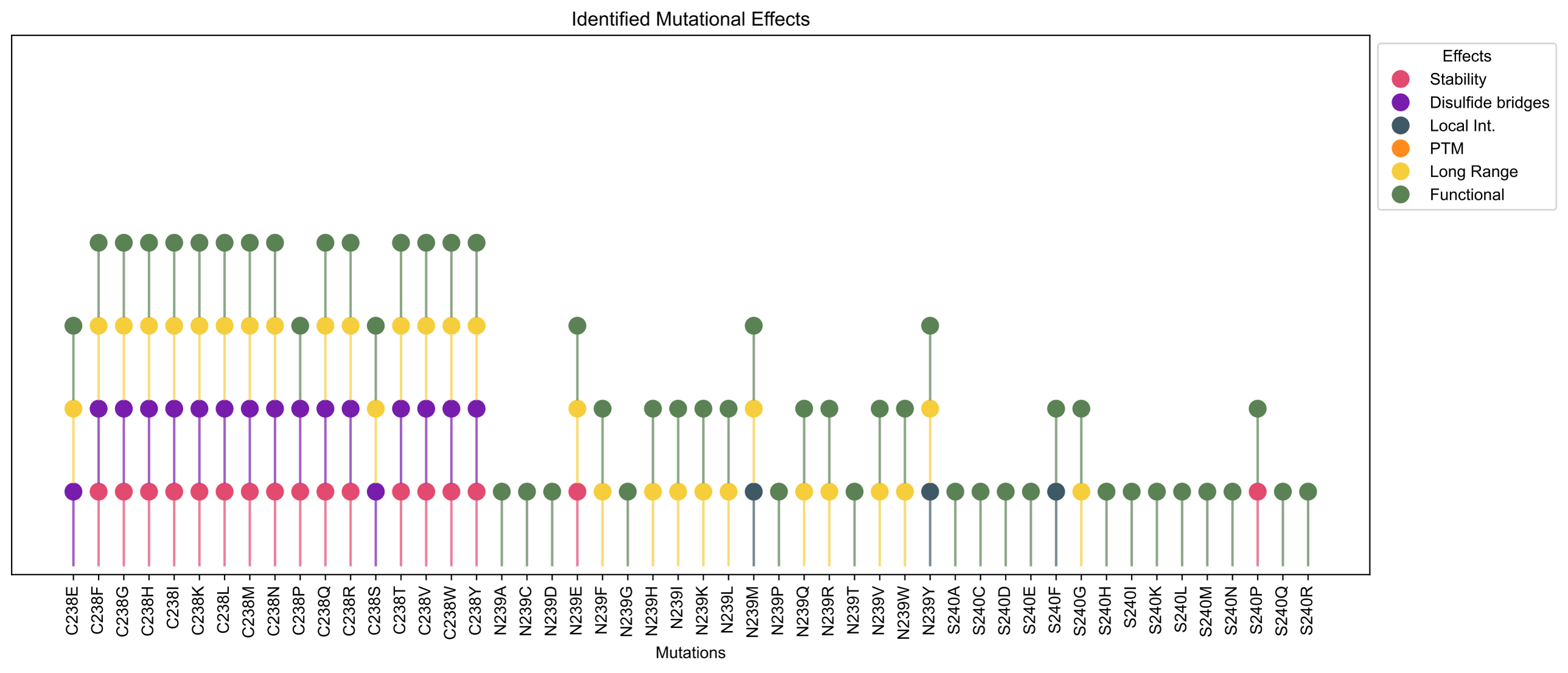

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
T211V
T211W
T211Y
F212P
R213A
R213C
R213D
R213E
R213F
R213G
R213H
R213I
R213L
R213M
R213N
R213P
R213Q
R213S
R213T
R213V
R213W
R213Y
H214I
H214L
H214W
S215A
S215G
S215I
S215K
S215L
S215P
S215T
S215V
V216A
V216C
V216D
V216E
V216F
V216G
V216H
V216K
V216N
V216P
V216Q
V216R
V216S
V216T
V216W
V216Y
V217D
Mutations

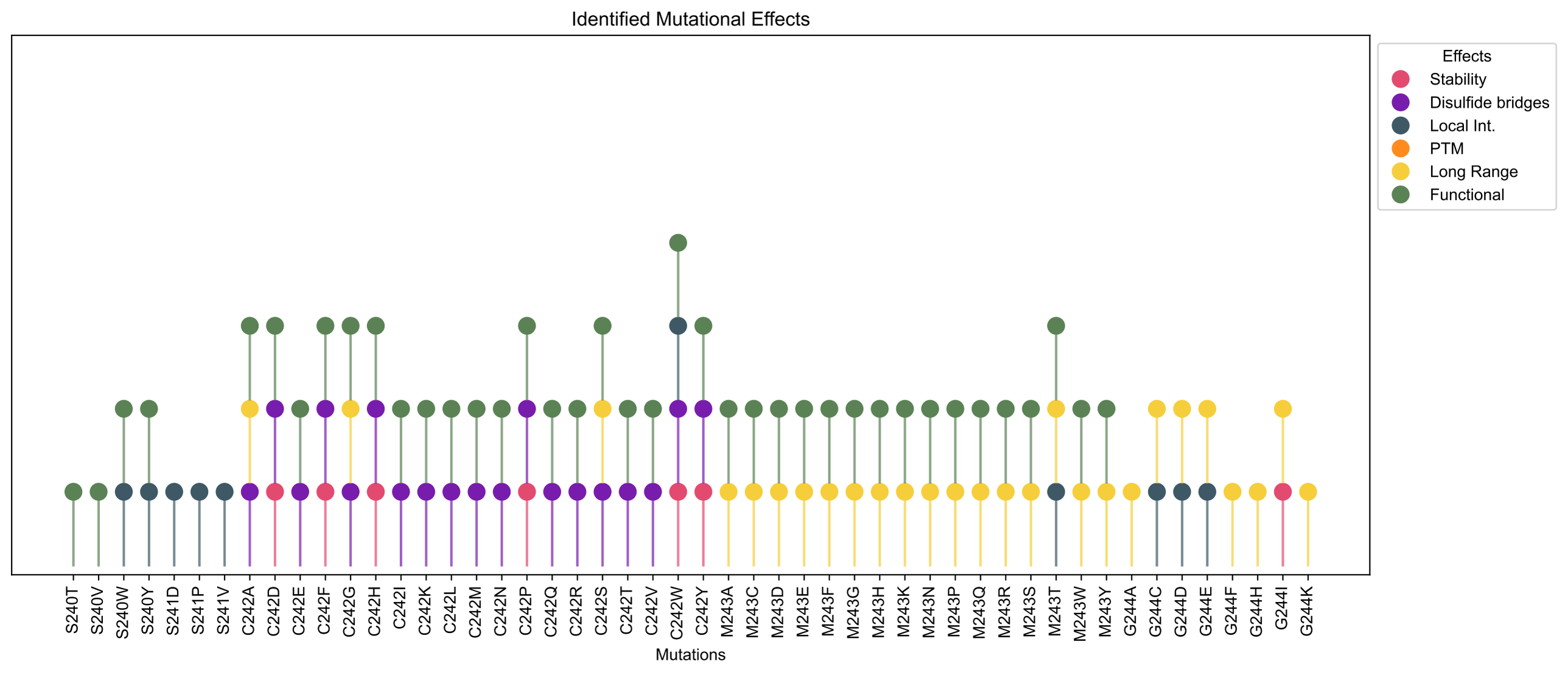

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
V217H
V217K
V217P
V217R
V218A
V218C
V218D
V218E
V218G
V218H
V218K
V218N
V218P
V218S
V218T
V218Y
P219G
Y220A
Y220C
Y220D
Y220E
Y220G
Y220H
Y220I
Y220K
Y220L
Y220M
Y220N
Y220P
Y220Q
Y220R
Y220S
Y220T
Y220V
Y220W
E221C
E221F
E221G
E221H
E221I
E221K
E221L
E221M
E221N
E221P
E221R
E221T
E221W
E221Y
P222W
Mutations

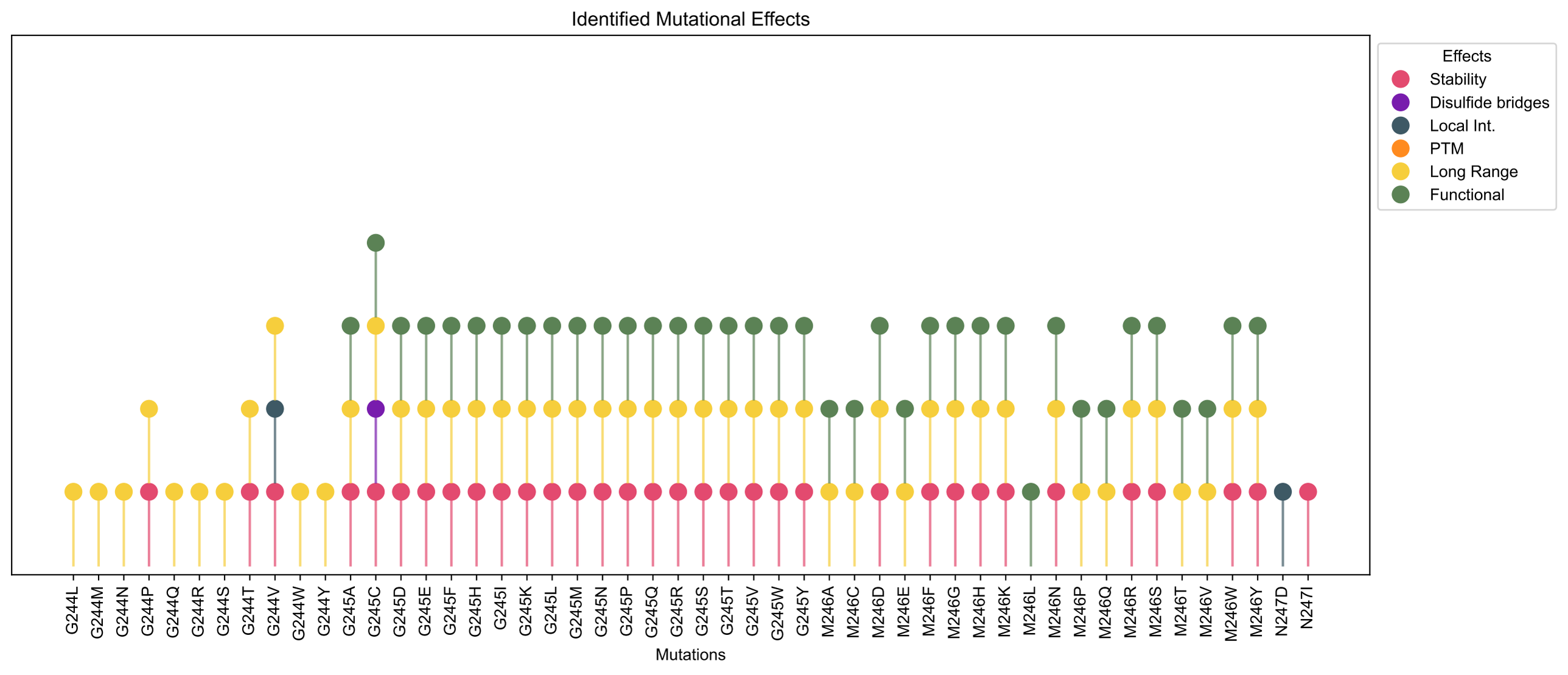

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
P223D
P223I
P223V
E224C
E224F
E224G
E224H
E224I
E224K
E224N
E224P
E224R
E224S
E224T
E224W
E224Y
V225C
V225D
V225H
V225K
V225N
V225R
V225W
V225Y
G226A
G226C
G226D
G226E
G226F
G226H
G226I
G226K
G226L
G226M
G226N
G226P
G226Q
G226R
G226S
G226T
G226V
G226W
G226Y
D228F
D228M
D228W
C229P
T230G
T230P
T231C
Mutations

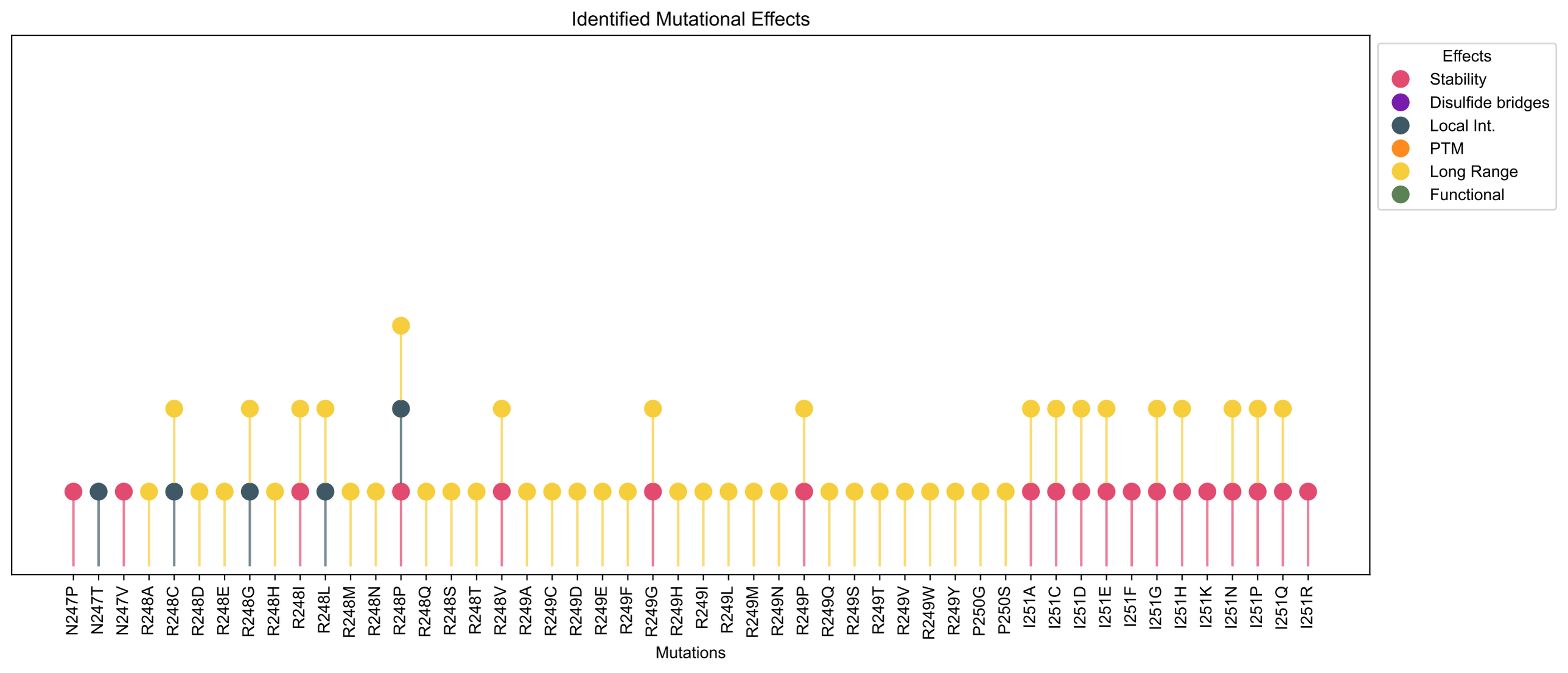

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
T231E
T231F
T231G
T231H
T231I
T231K
T231Q
T231R
T231W
T231Y
I232A
I232C
I232D
I232E
I232G
I232H
I232N
I232P
I232Q
I232R
I232S
I232W
I232Y
H233C
H233D
H233E
H233G
H233I
H233K
H233P
H233R
H233T
H233V
H233W
Y234A
Y234C
Y234D
Y234E
Y234G
Y234H
Y234I
Y234K
Y234M
Y234N
Y234P
Y234Q
Y234R
Y234S
Y234T
Y234V
Mutations

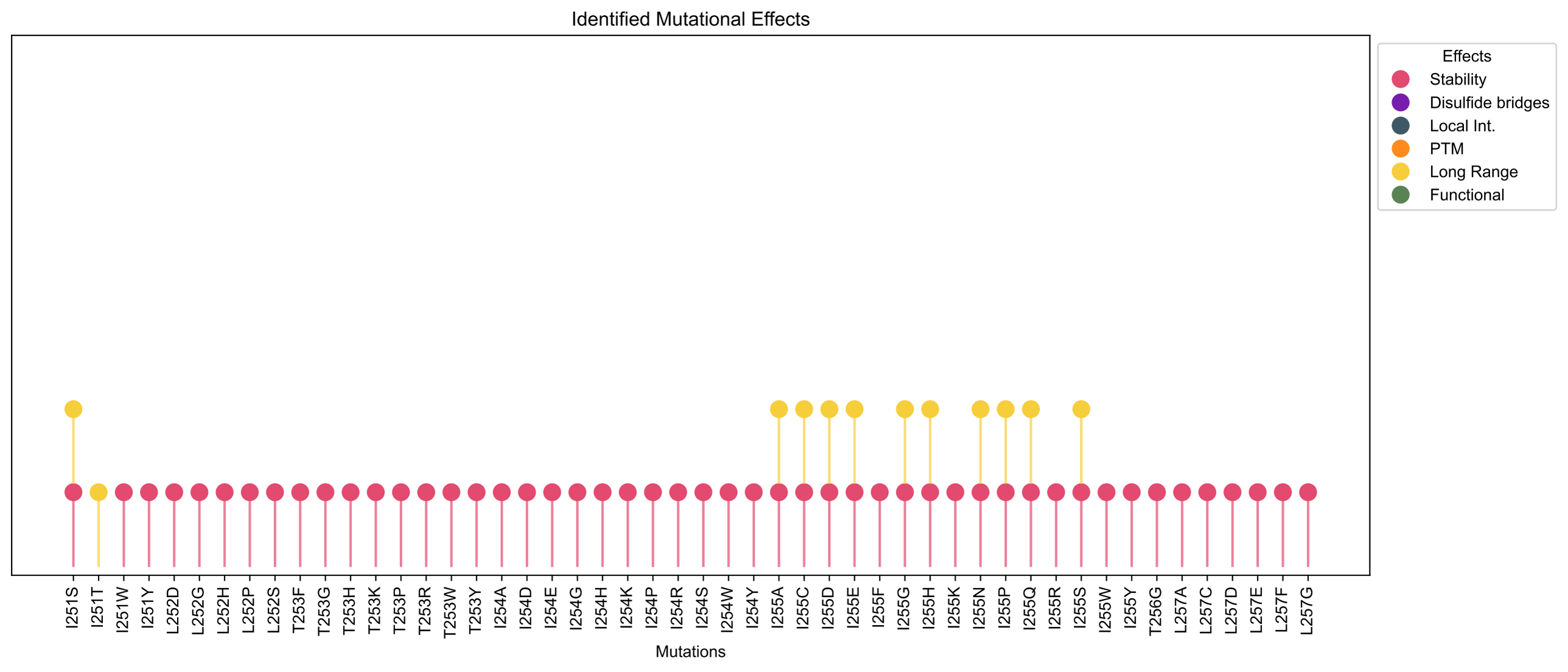

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
Y234W
N235A
N235E
N235F
N235G
N235I
N235L
N235M
N235P
N235Q
N235V
N235W
N235Y
Y236A
Y236C
Y236D
Y236E
Y236G
Y236H
Y236I
Y236K
Y236L
Y236M
Y236N
Y236P
Y236Q
Y236R
Y236S
Y236T
Y236V
Y236W
M237A
M237C
M237D
M237E
M237F
M237G
M237H
M237K
M237N
M237P
M237Q
M237R
M237S
M237T
M237V
M237W
M237Y
C238A
C238D
Mutations

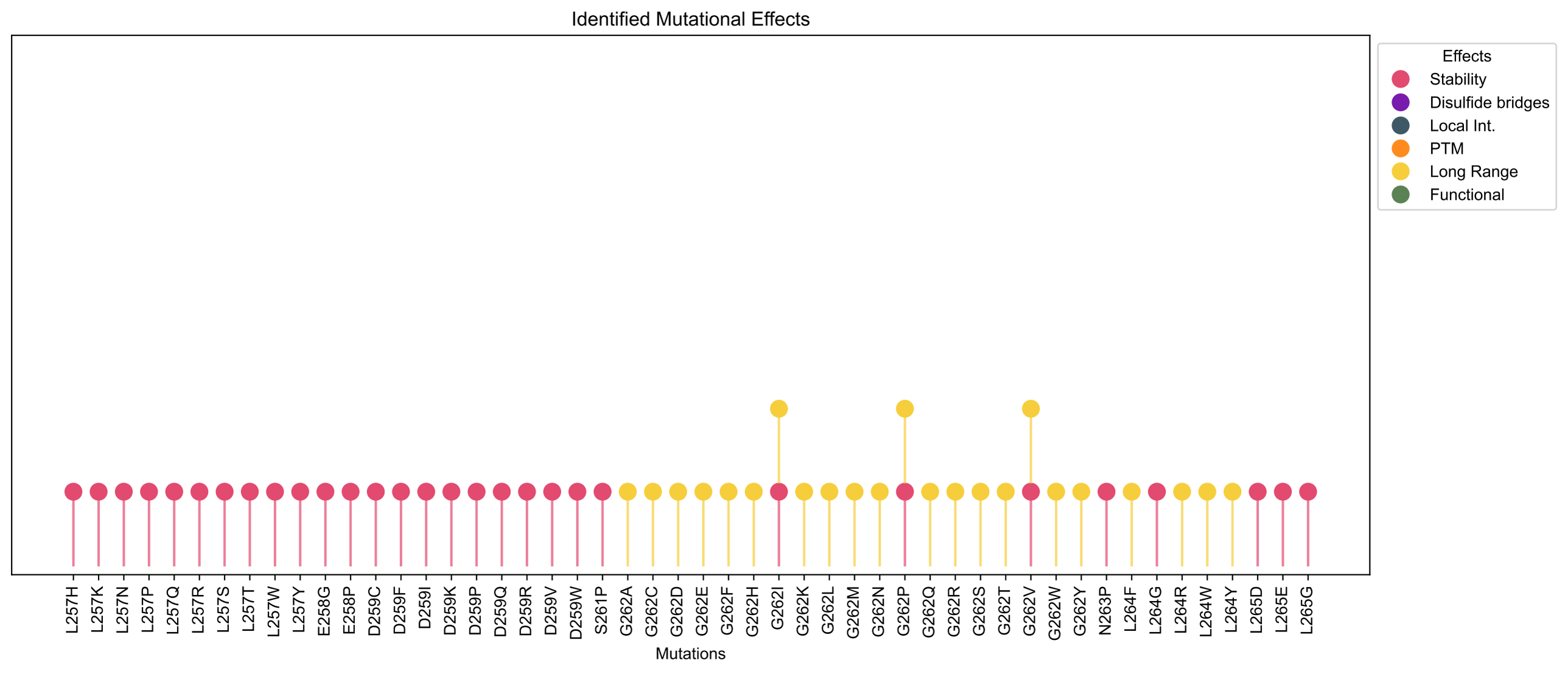

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
C238E
C238F
C238G
C238H
C238I
C238K
C238L
C238M
C238N
C238P
C238Q
C238R
C238S
C238T
C238V
C238W
C238Y
N239A
N239C
N239D
N239E
N239F
N239G
N239H
N239I
N239K
N239L
N239M
N239P
N239Q
N239R
N239T
N239V
N239W
N239Y
S240A
S240C
S240D
S240E
S240F
S240G
S240H
S240I
S240K
S240L
S240M
S240N
S240P
S240Q
S240R
Mutations

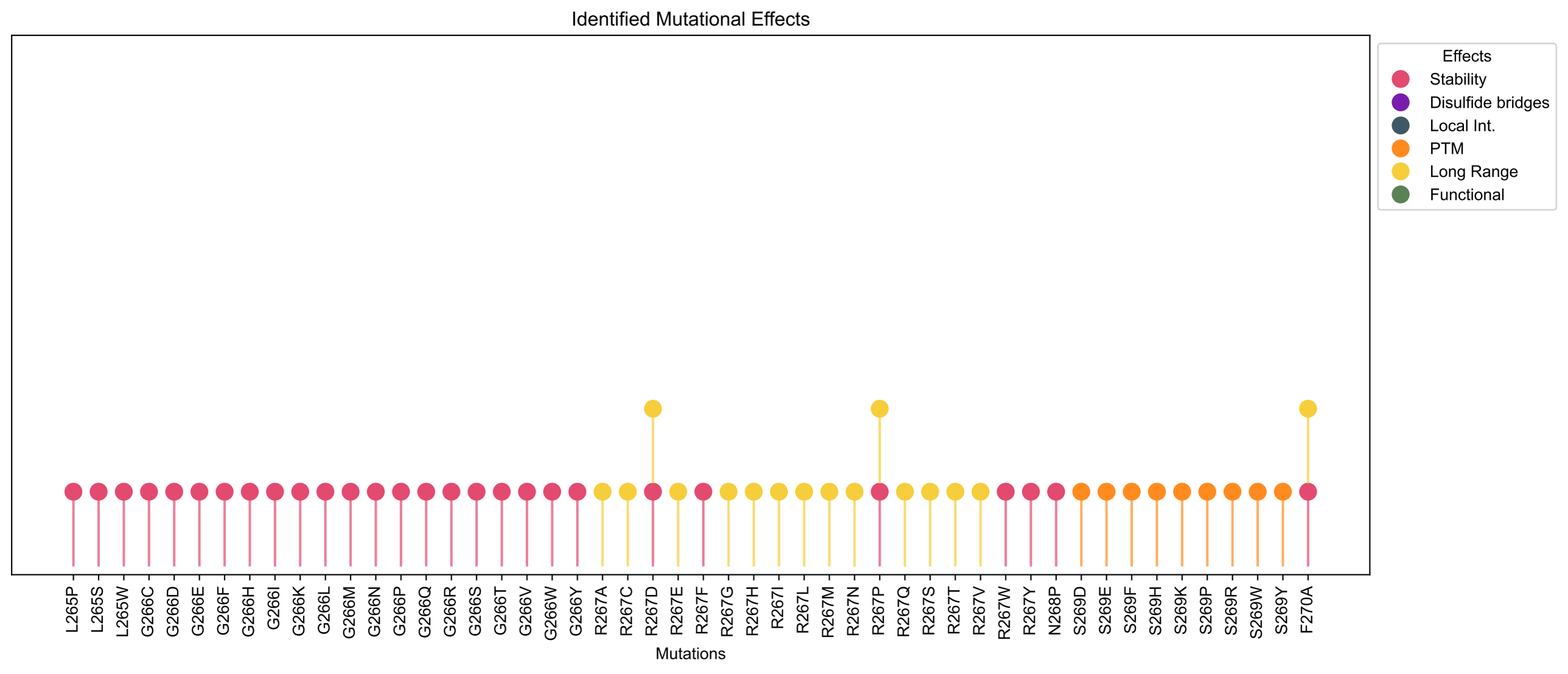

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
S240T
S240V
S240W
S240Y
S241D
S241P
S241V
C242A
C242D
C242E
C242F
C242G
C242H
C242I
C242K
C242L
C242M
C242N
C242P
C242Q
C242R
C242S
C242T
C242V
C242W
C242Y
M243A
M243C
M243D
M243E
M243F
M243G
M243H
M243K
M243N
M243P
M243Q
M243R
M243S
M243T
M243W
M243Y
G244A
G244C
G244D
G244E
G244F
G244H
G244I
G244K
Mutations

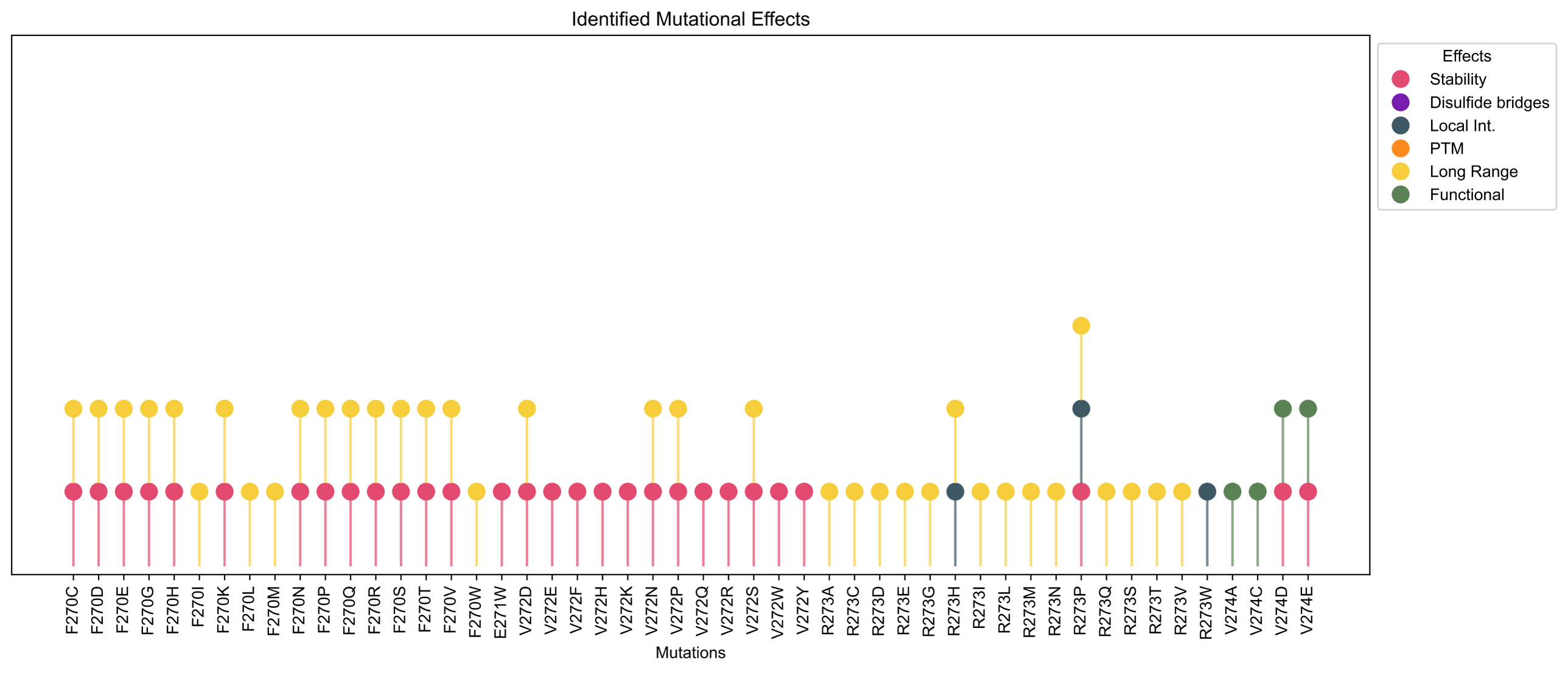

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
G244L
G244M
G244N
G244P
G244Q
G244R
G244S
G244T
G244V
G244W
G244Y
G245A
G245C
G245D
G245E
G245F
G245H
G245I
G245K
G245L
G245M
G245N
G245P
G245Q
G245R
G245S
G245T
G245V
G245W
G245Y
M246A
M246C
M246D
M246E
M246F
M246G
M246H
M246K
M246L
M246N
M246P
M246Q
M246R
M246S
M246T
M246V
M246W
M246Y
N247D
N247I
Mutations

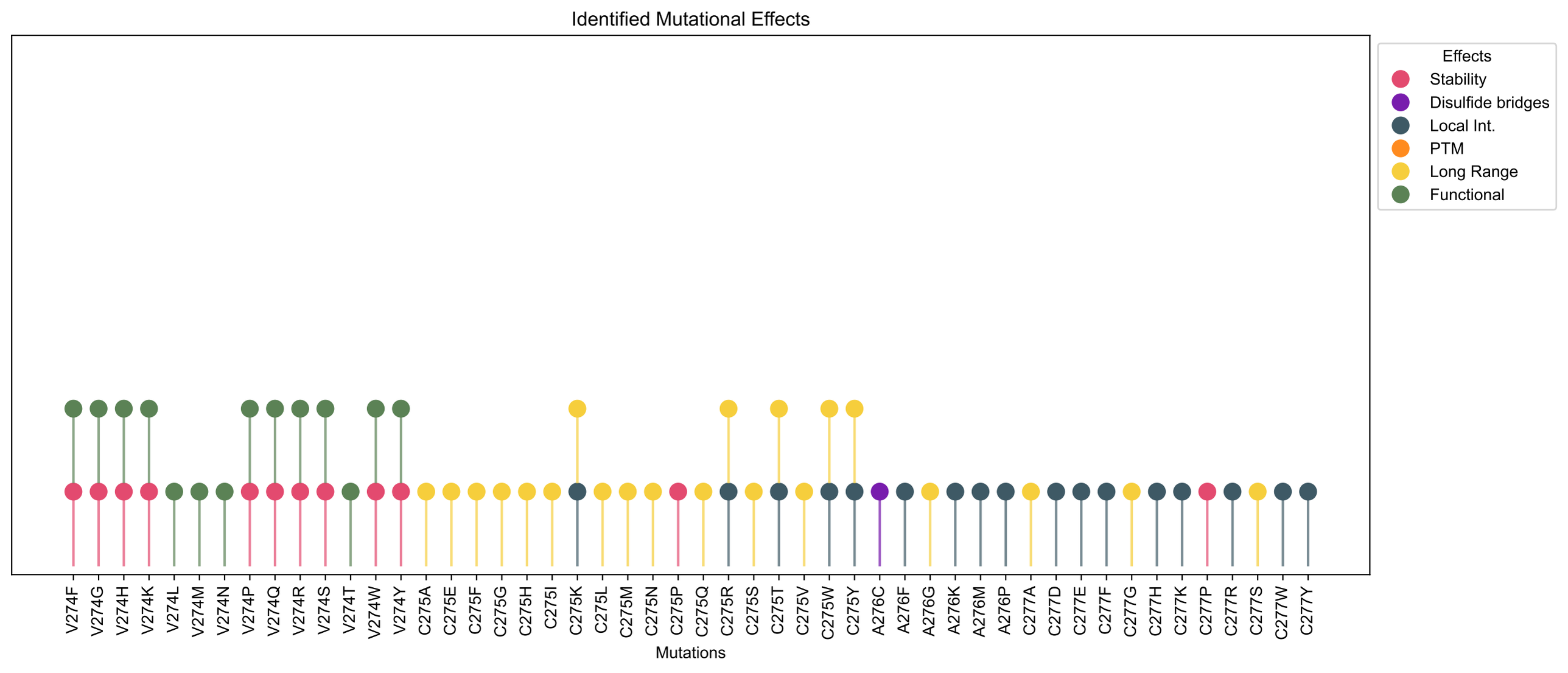

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
N247P
N247T
N247V
R248A
R248C
R248D
R248E
R248G
R248H
R248I
R248L
R248M
R248N
R248P
R248Q
R248S
R248T
R248V
R249A
R249C
R249D
R249E
R249F
R249G
R249H
R249I
R249L
R249M
R249N
R249P
R249Q
R249S
R249T
R249V
R249W
R249Y
P250G
P250S
I251A
I251C
I251D
I251E
I251F
I251G
I251H
I251K
I251N
I251P
I251Q
I251R
Mutations

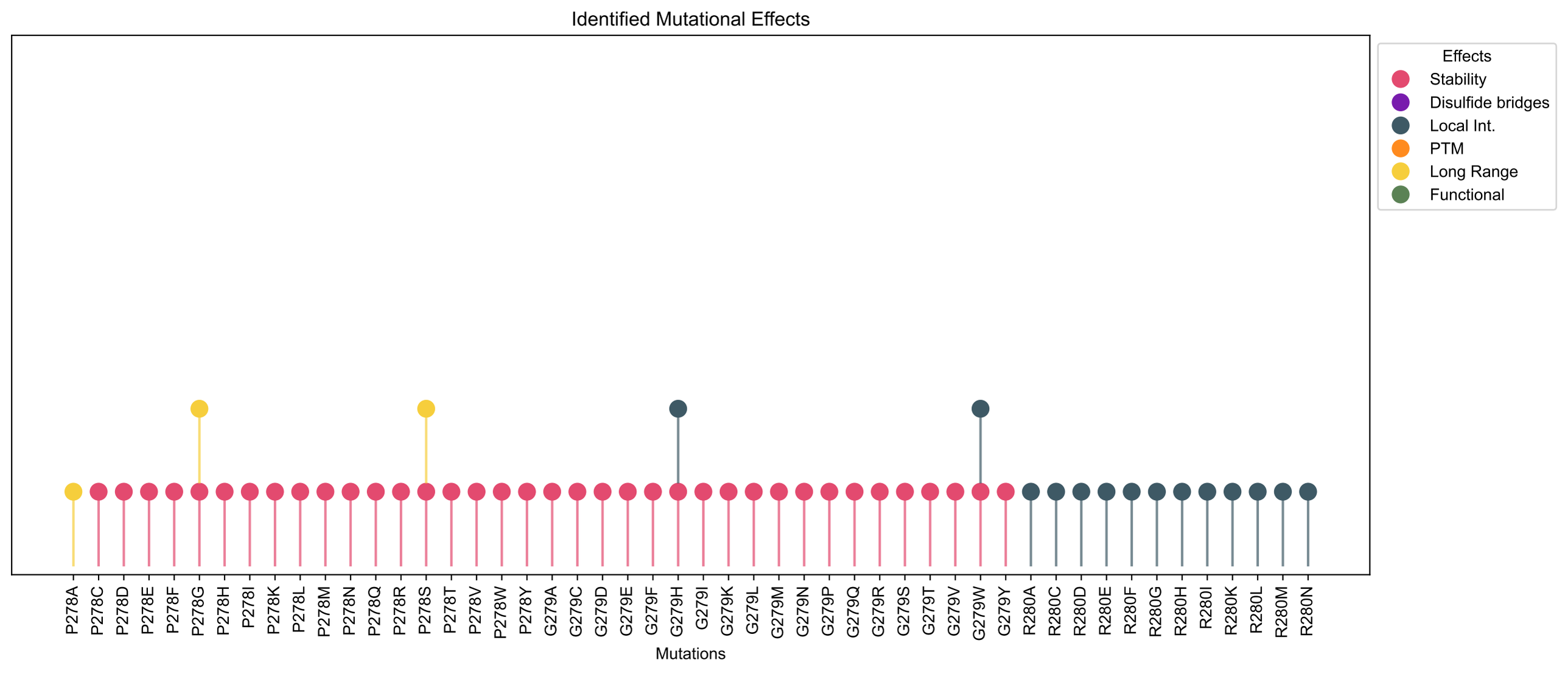

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
I251S
I251T
I251W
I251Y
L252D
L252G
L252H
L252P
L252S
T253F
T253G
T253H
T253K
T253P
T253R
T253W
T253Y
I254A
I254D
I254E
I254G
I254H
I254K
I254P
I254R
I254S
I254W
I254Y
I255A
I255C
I255D
I255E
I255F
I255G
I255H
I255K
I255N
I255P
I255Q
I255R
I255S
I255W
I255Y
T256G
L257A
L257C
L257D
L257E
L257F
L257G
Mutations

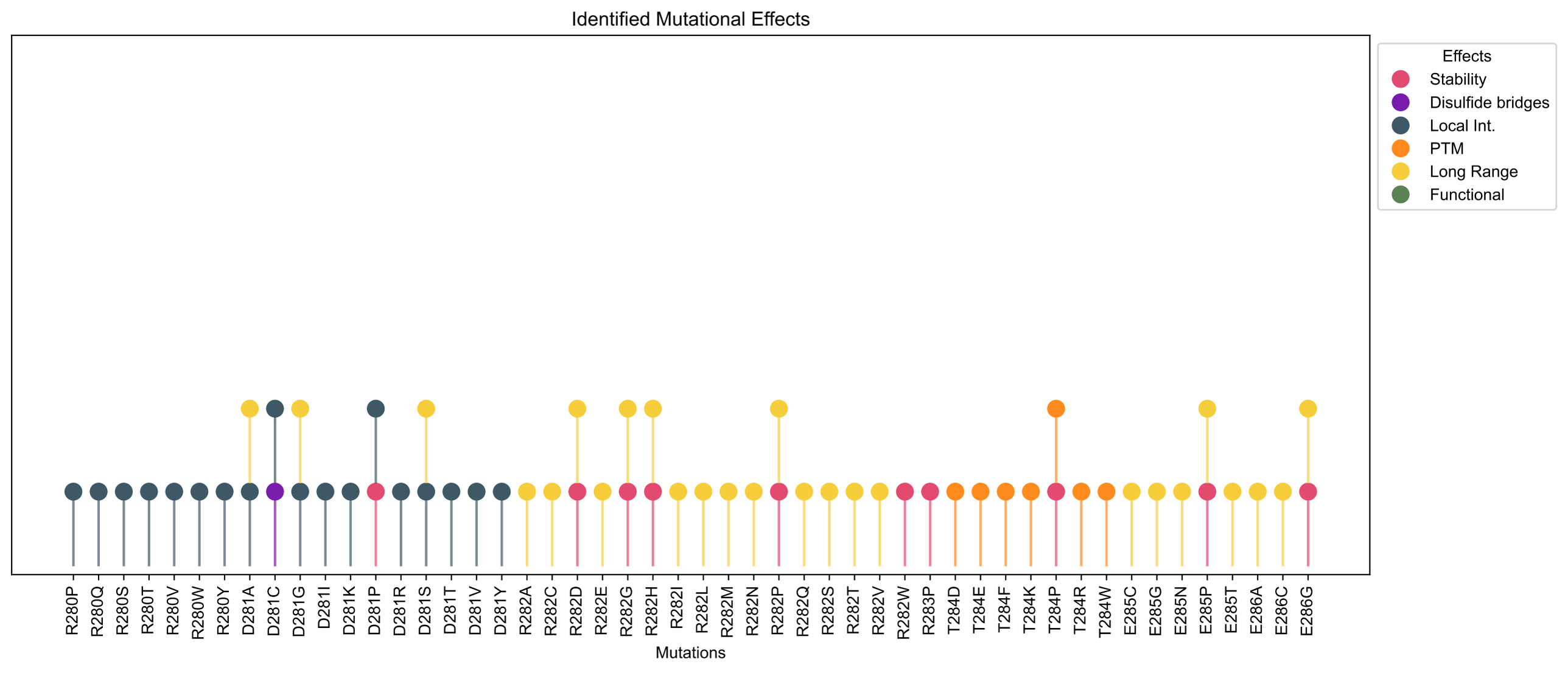

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
L257H
L257K
L257N
L257P
L257Q
L257R
L257S
L257T
L257W
L257Y
E258G
E258P
D259C
D259F
D259I
D259K
D259P
D259Q
D259R
D259V
D259W
S261P
G262A
G262C
G262D
G262E
G262F
G262H
G262I
G262K
G262L
G262M
G262N
G262P
G262Q
G262R
G262S
G262T
G262V
G262W
G262Y
N263P
L264F
L264G
L264R
L264W
L264Y
L265D
L265E
L265G
Mutations

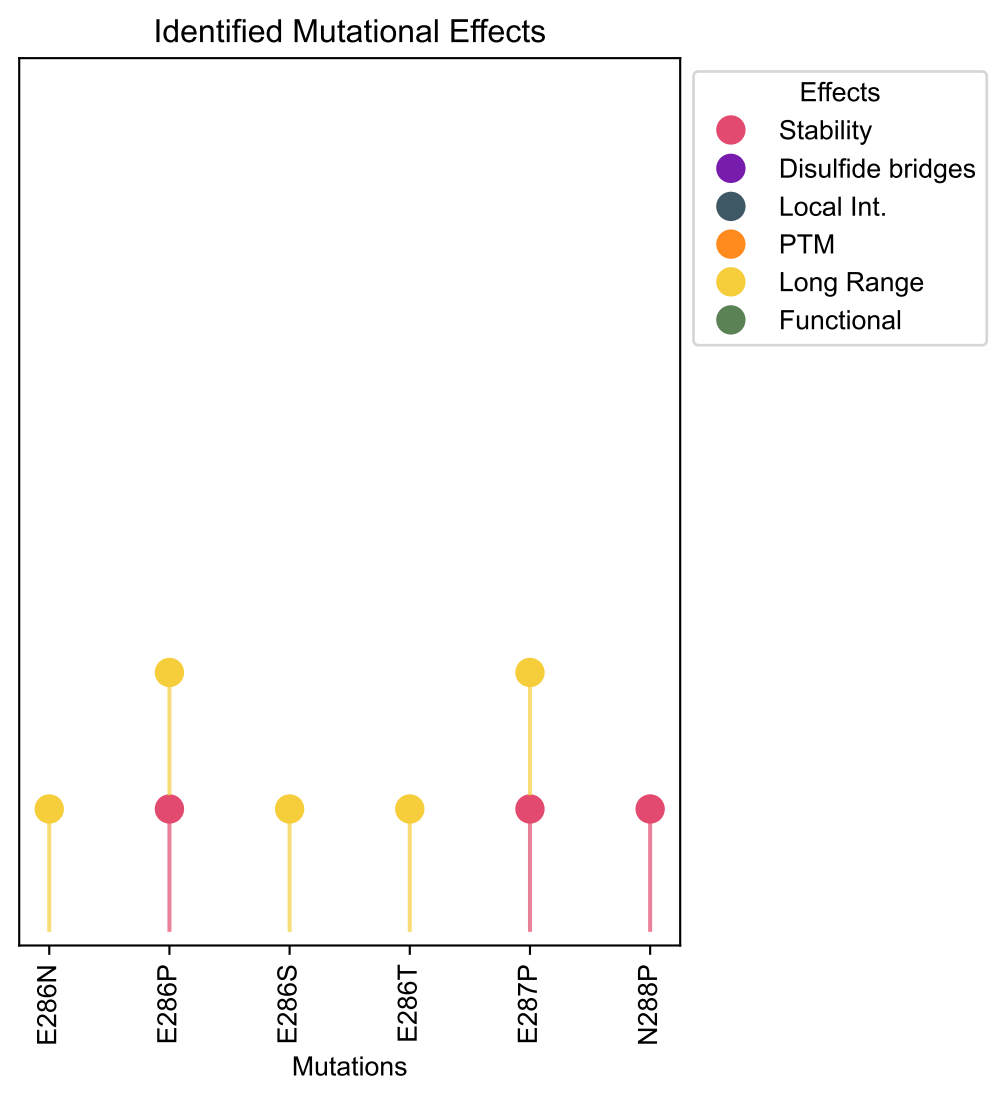

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
L265P
L265S
L265W
G266C
G266D
G266E
G266F
G266H
G266I
G266K
G266L
G266M
G266N
G266P
G266Q
G266R
G266S
G266T
G266V
G266W
G266Y
R267A
R267C
R267D
R267E
R267F
R267G
R267H
R267I
R267L
R267M
R267N
R267P
R267Q
R267S
R267T
R267V
R267W
R267Y
N268P
S269D
S269E
S269F
S269H
S269K
S269P
S269R
S269W
S269Y
F270A
Mutations


Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
F270C
F270D
F270E
F270G
F270H
F270I
F270K
F270L
F270M
F270N
F270P
F270Q
F270R
F270S
F270T
F270V
F270W
E271W
V272D
V272E
V272F
V272H
V272K
V272N
V272P
V272Q
V272R
V272S
V272W
V272Y
R273A
R273C
R273D
R273E
R273G
R273H
R273I
R273L
R273M
R273N
R273P
R273Q
R273S
R273T
R273V
R273W
V274A
V274C
V274D
V274E
Mutations


Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
V274F
V274G
V274H
V274K
V274L
V274M
V274N
V274P
V274Q
V274R
V274S
V274T
V274W
V274Y
C275A
C275E
C275F
C275G
C275H
C275I
C275K
C275L
C275M
C275N
C275P
C275Q
C275R
C275S
C275T
C275V
C275W
C275Y
A276C
A276F
A276G
A276K
A276M
A276P
C277A
C277D
C277E
C277F
C277G
C277H
C277K
C277P
C277R
C277S
C277W
C277Y
Mutations


Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
P278A
P278C
P278D
P278E
P278F
P278G
P278H
P278I
P278K
P278L
P278M
P278N
P278Q
P278R
P278S
P278T
P278V
P278W
P278Y
G279A
G279C
G279D
G279E
G279F
G279H
G279I
G279K
G279L
G279M
G279N
G279P
G279Q
G279R
G279S
G279T
G279V
G279W
G279Y
R280A
R280C
R280D
R280E
R280F
R280G
R280H
R280I
R280K
R280L
R280M
R280N
Mutations


Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
R280P
R280Q
R280S
R280T
R280V
R280W
R280Y
D281A
D281C
D281G
D281I
D281K
D281P
D281R
D281S
D281T
D281V
D281Y
R282A
R282C
R282D
R282E
R282G
R282H
R282I
R282L
R282M
R282N
R282P
R282Q
R282S
R282T
R282V
R282W
R283P
T284D
T284E
T284F
T284K
T284P
T284R
T284W
E285C
E285G
E285N
E285P
E285T
E286A
E286C
E286G
Mutations


Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
E286N
E286P
E286S
E286T
E287P
N288P
Mutations